

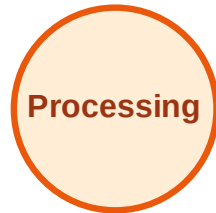
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



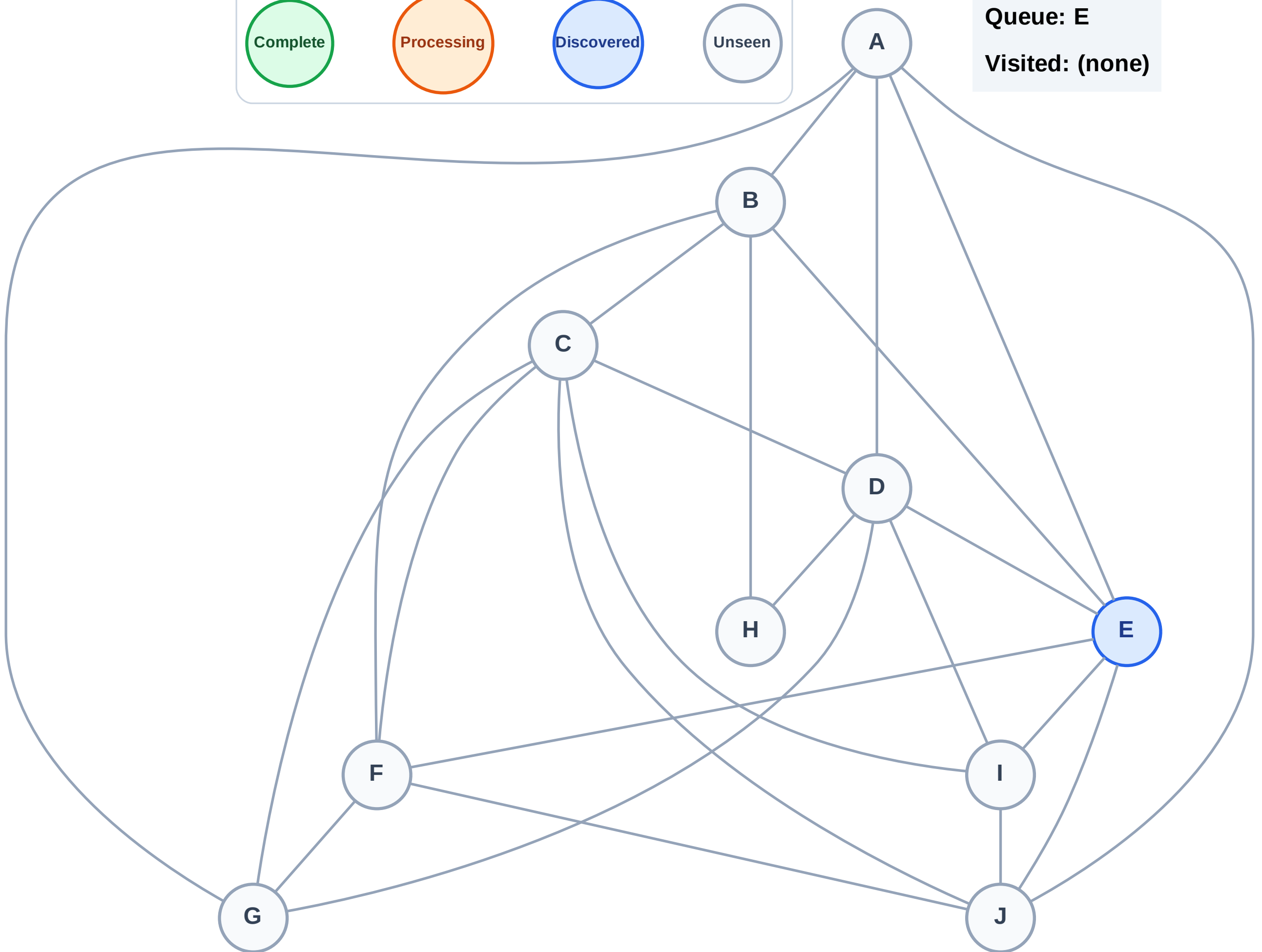
Discovered



Unseen

Queue: E

Visited: (none)



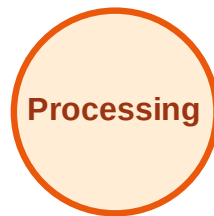
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



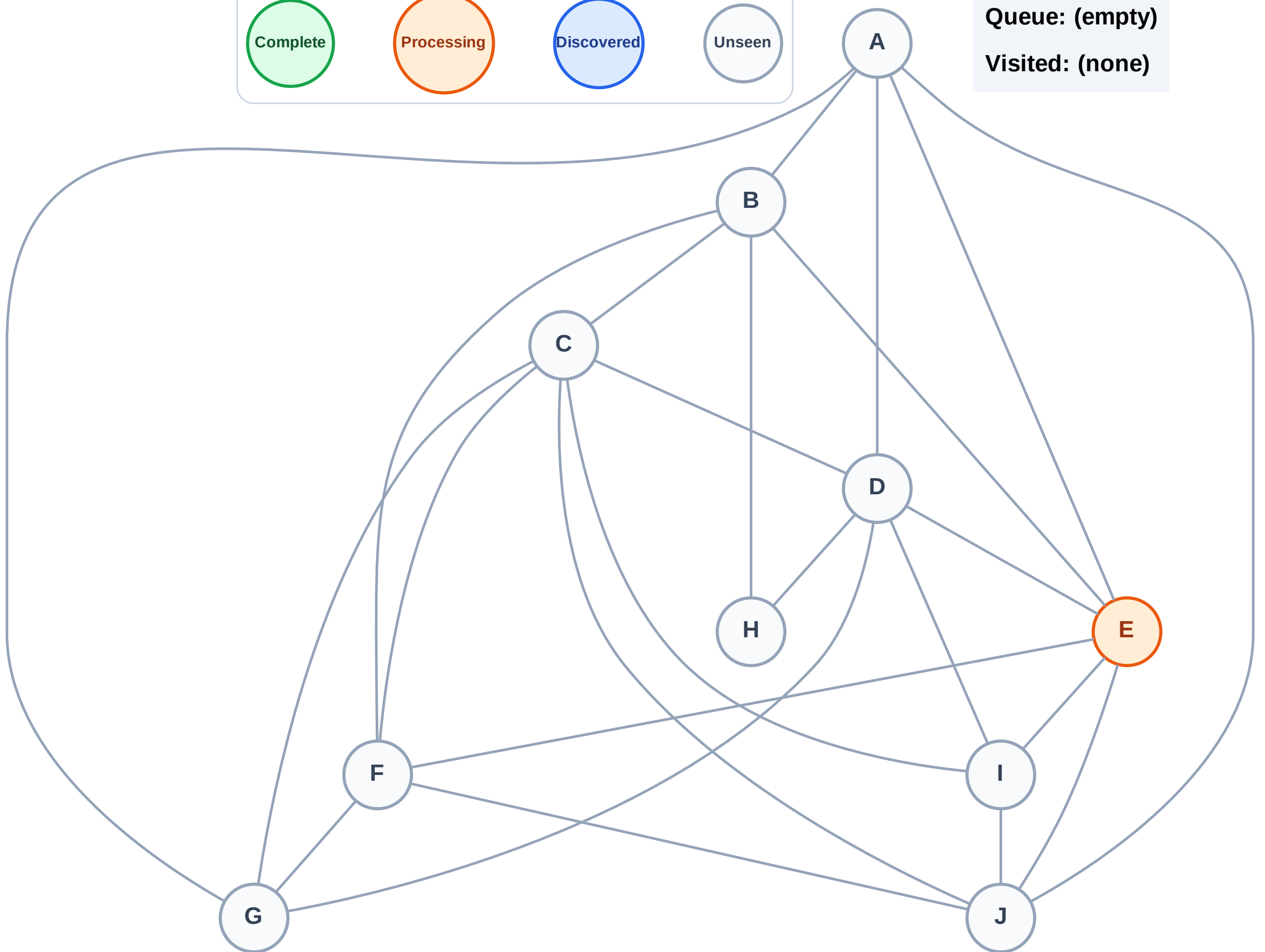
Discovered



Unseen

Queue: (empty)

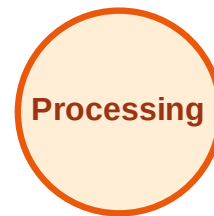
Visited: (none)



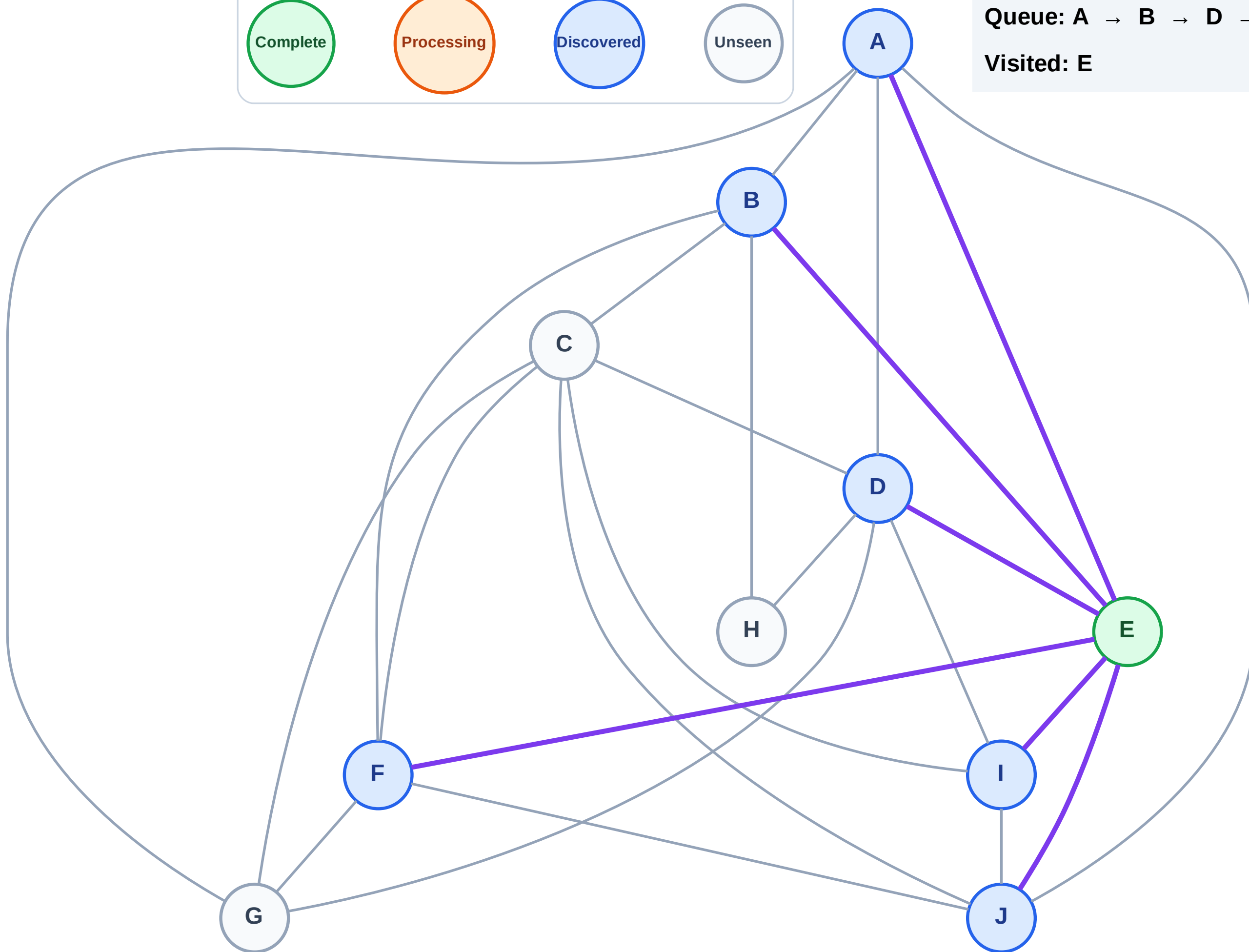
BFS Traversal

Step 3: Discover A, B, D, F, I, J from E

Legend



Queue: A → B → D → F → I → J
Visited: E



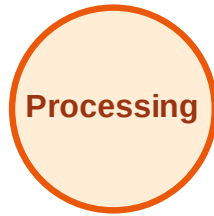
BFS Traversal

Step 4: Process node A

Legend



Complete



Processing

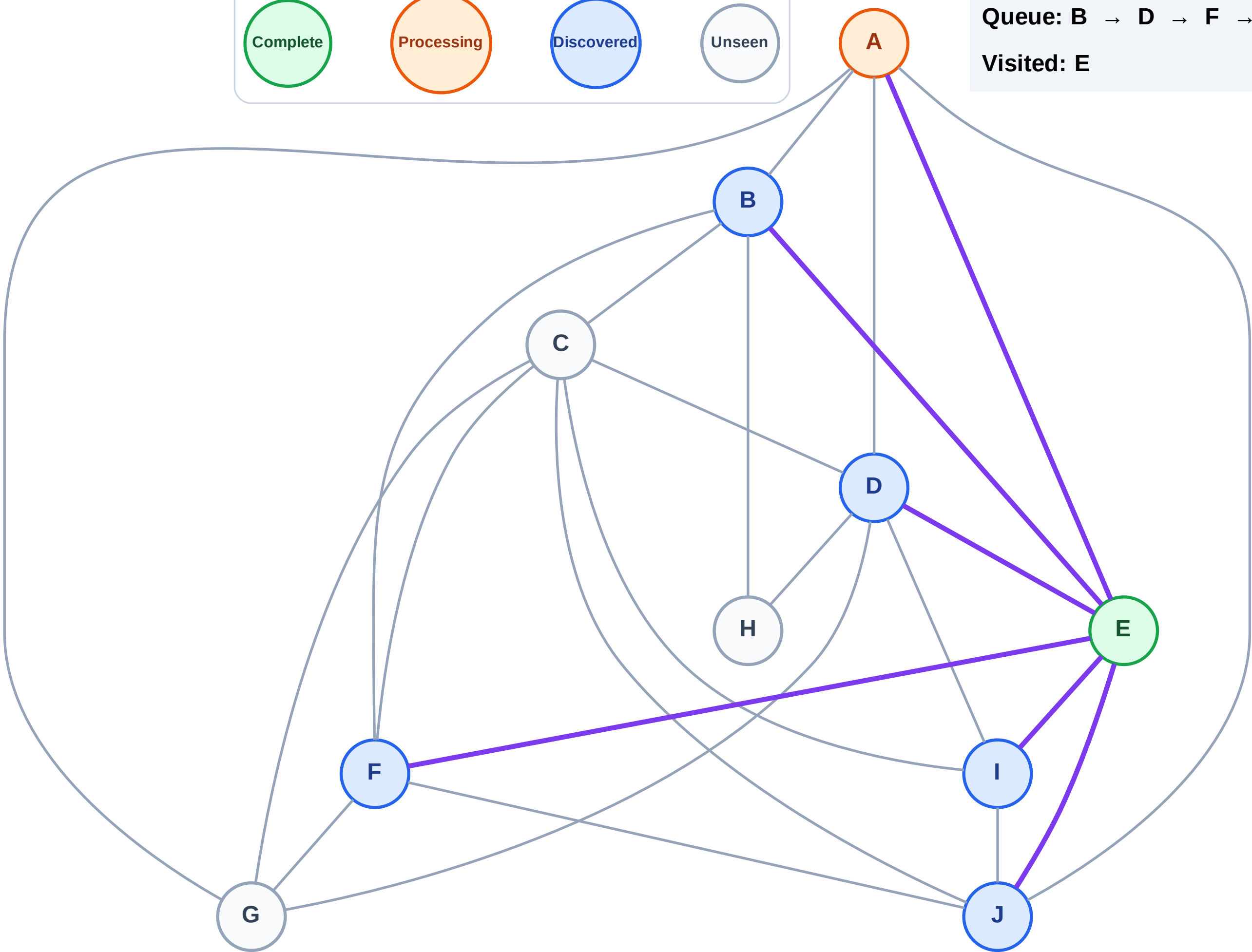


Discovered



Unseen

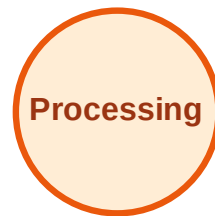
Queue: B → D → F → I → J
Visited: E



BFS Traversal

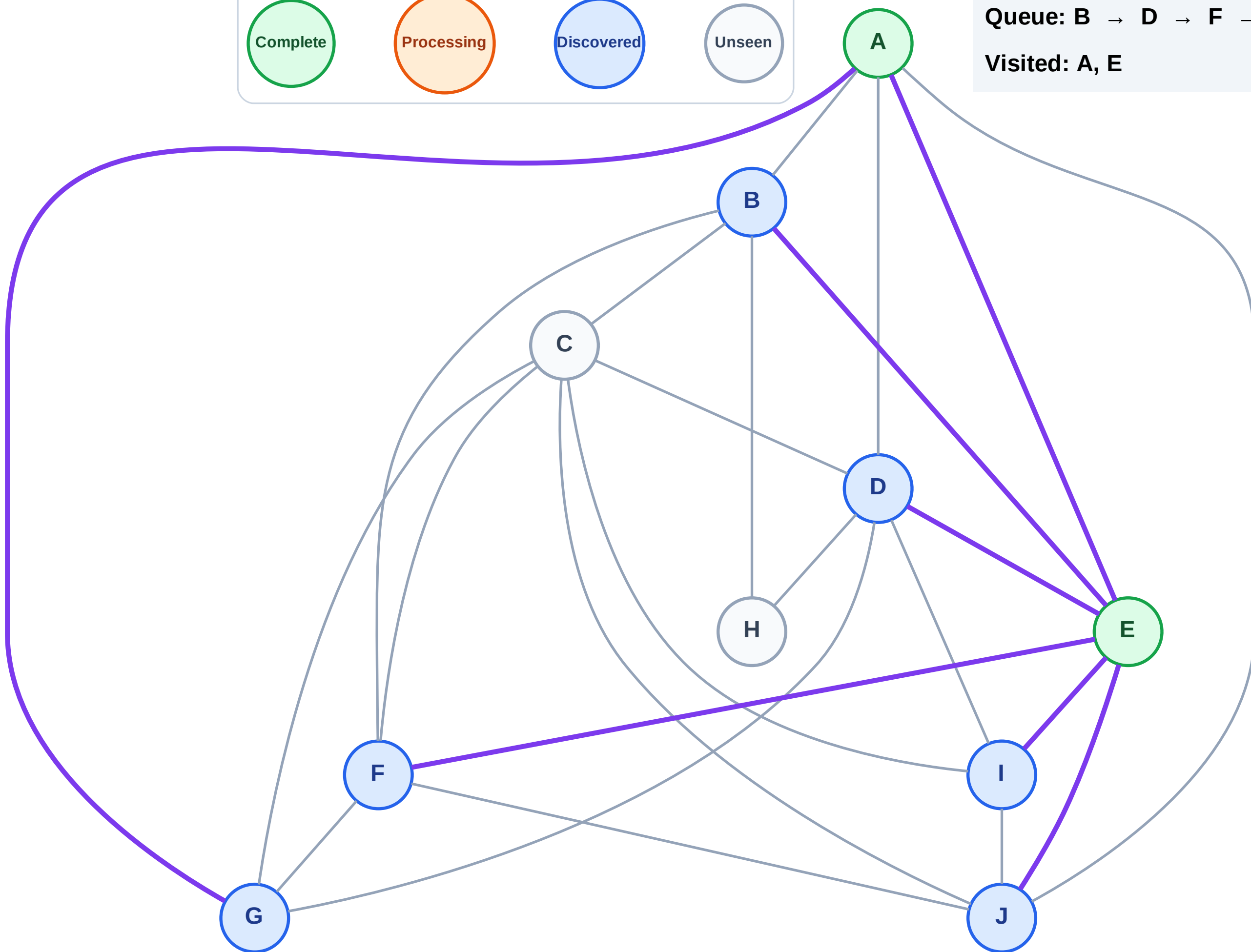
Step 5: Discover G from A

Legend



Queue: B → D → F → I → J → G

Visited: A, E



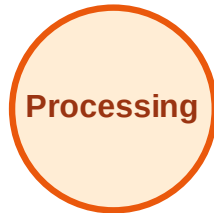
BFS Traversal

Step 6: Process node B

Legend



Complete



Processing

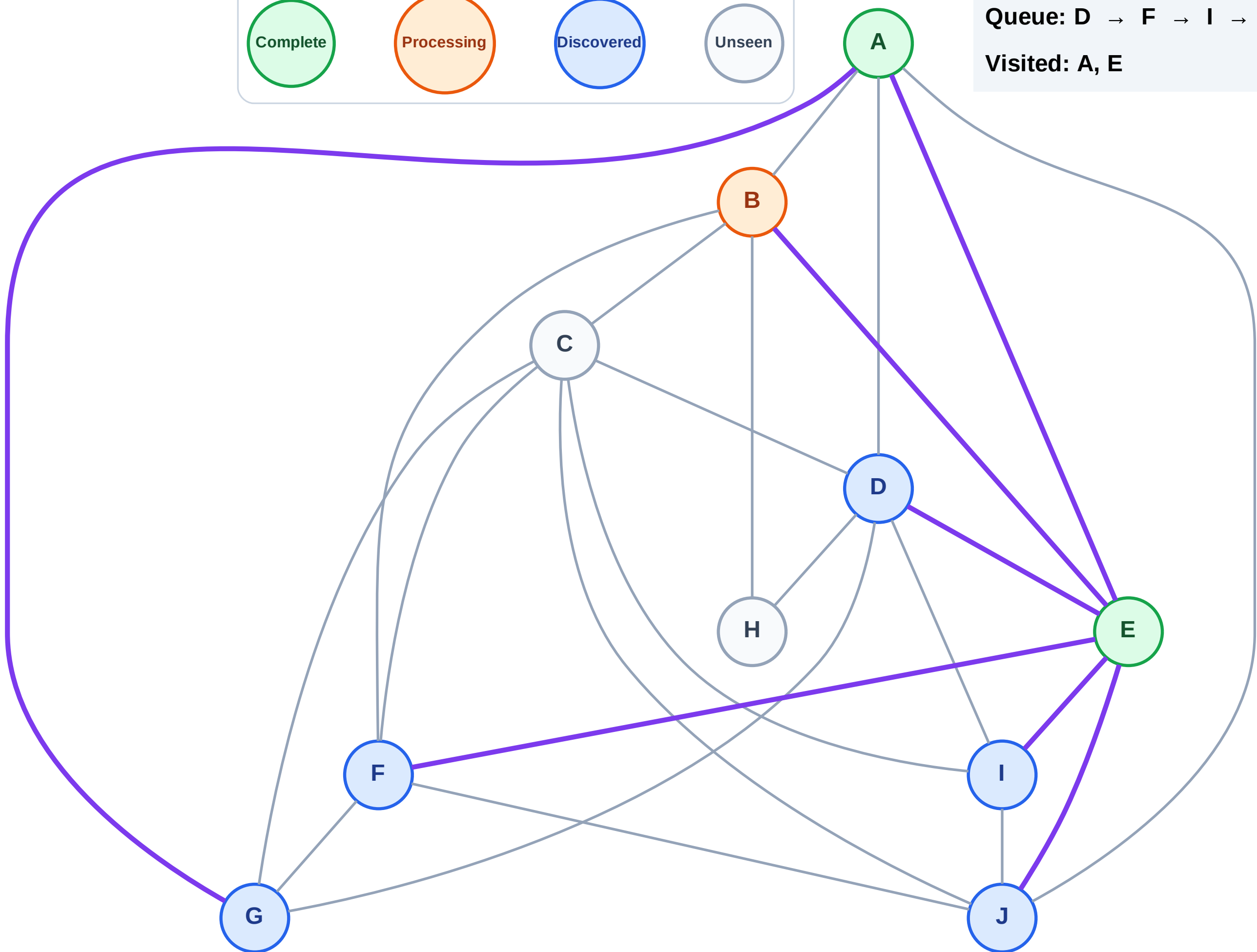


Discovered



Unseen

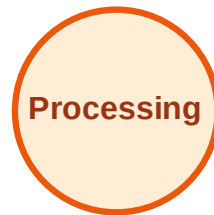
Queue: D → F → I → J → G
Visited: A, E



BFS Traversal

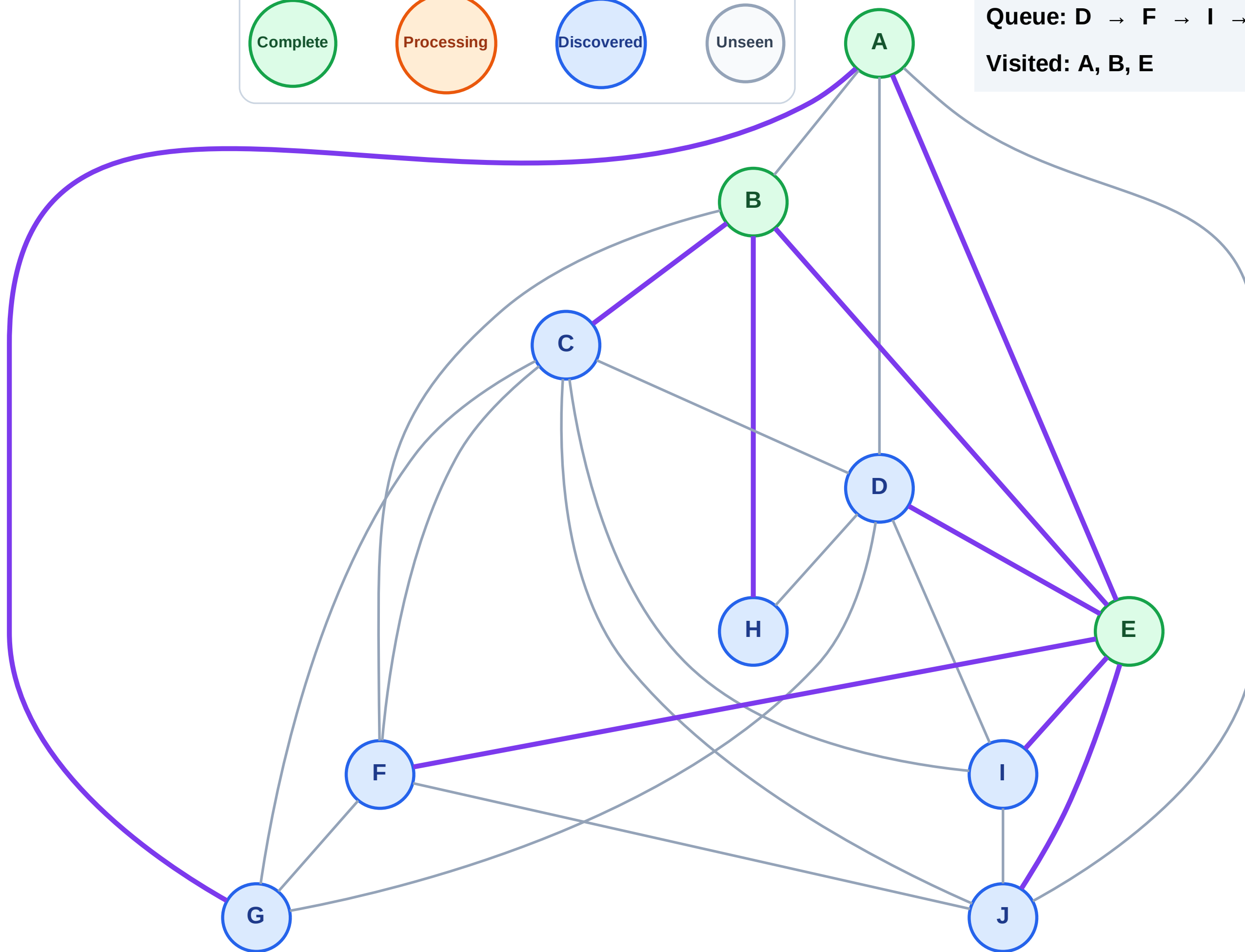
Step 7: Discover C, H from B

Legend



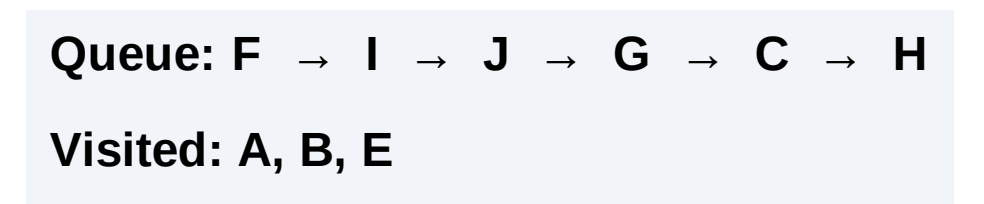
Queue: D → F → I → J → G → C → H

Visited: A, B, E



Step 8: Process node D

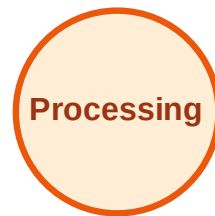
Step 8: Process node D



BFS Traversal

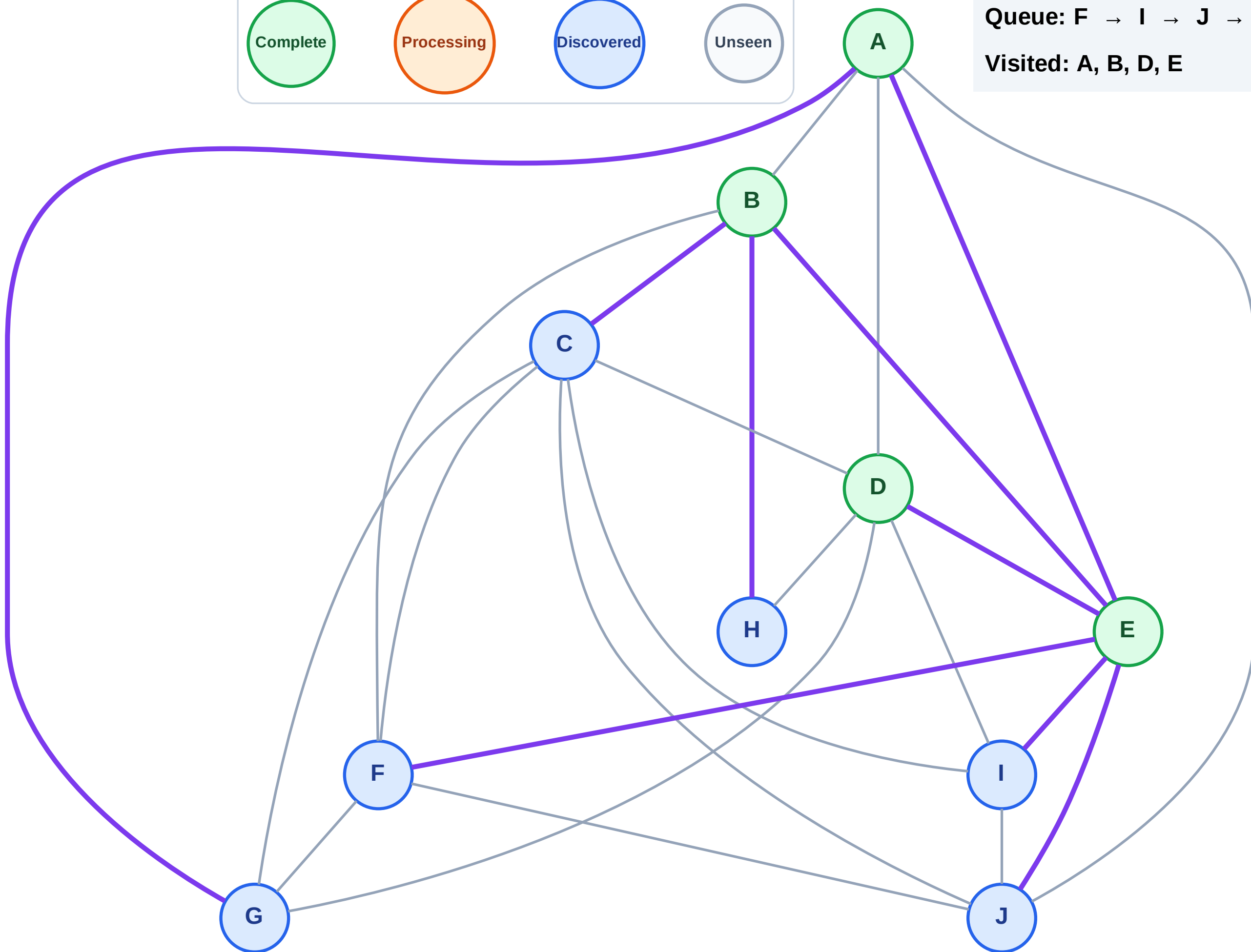
Step 9: No new neighbors from D

Legend



Queue: F → I → J → G → C → H

Visited: A, B, D, E



BFS Traversal

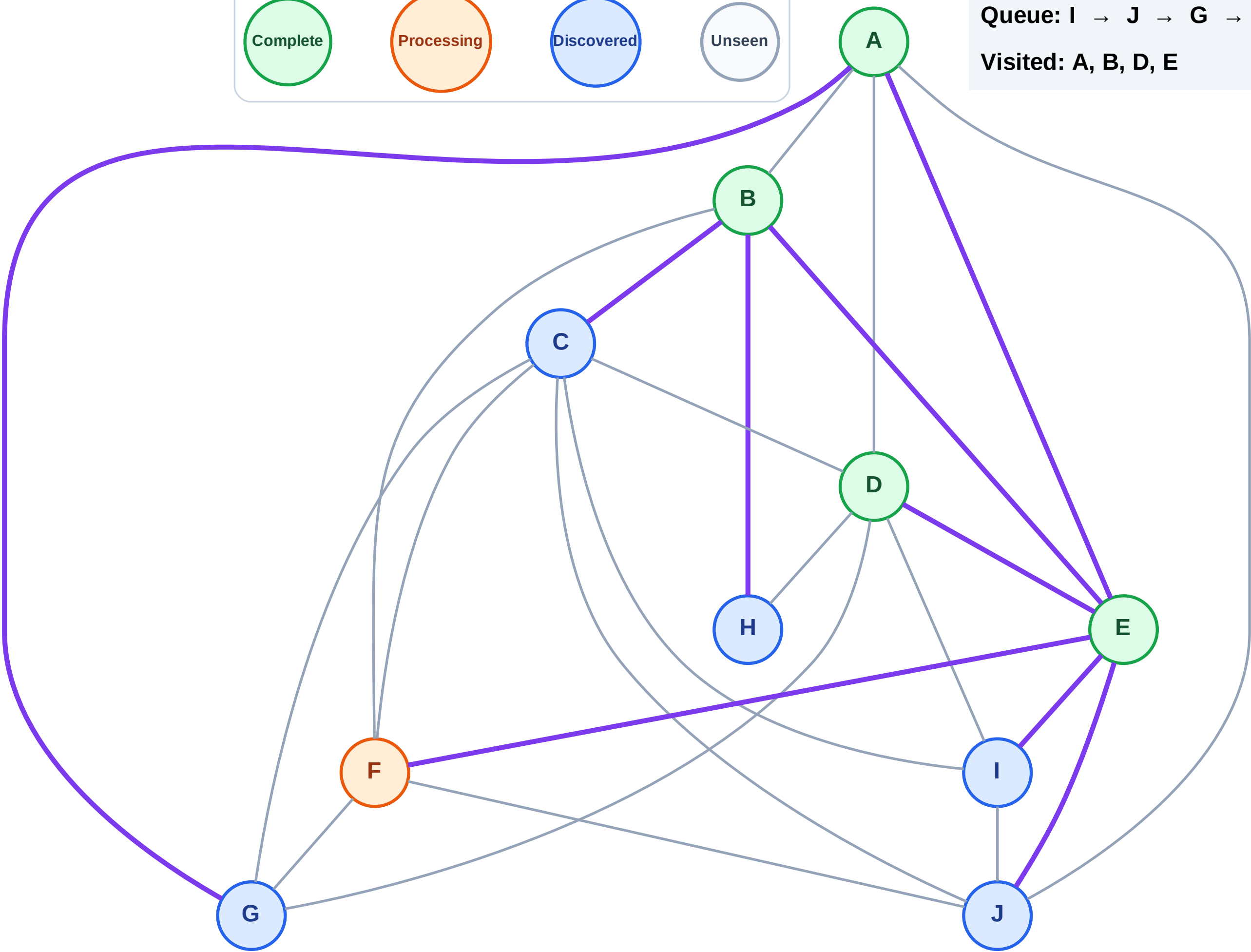
Step 10: Process node F

Legend



Queue: I → J → G → C → H

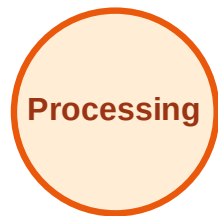
Visited: A, B, D, E



BFS Traversal

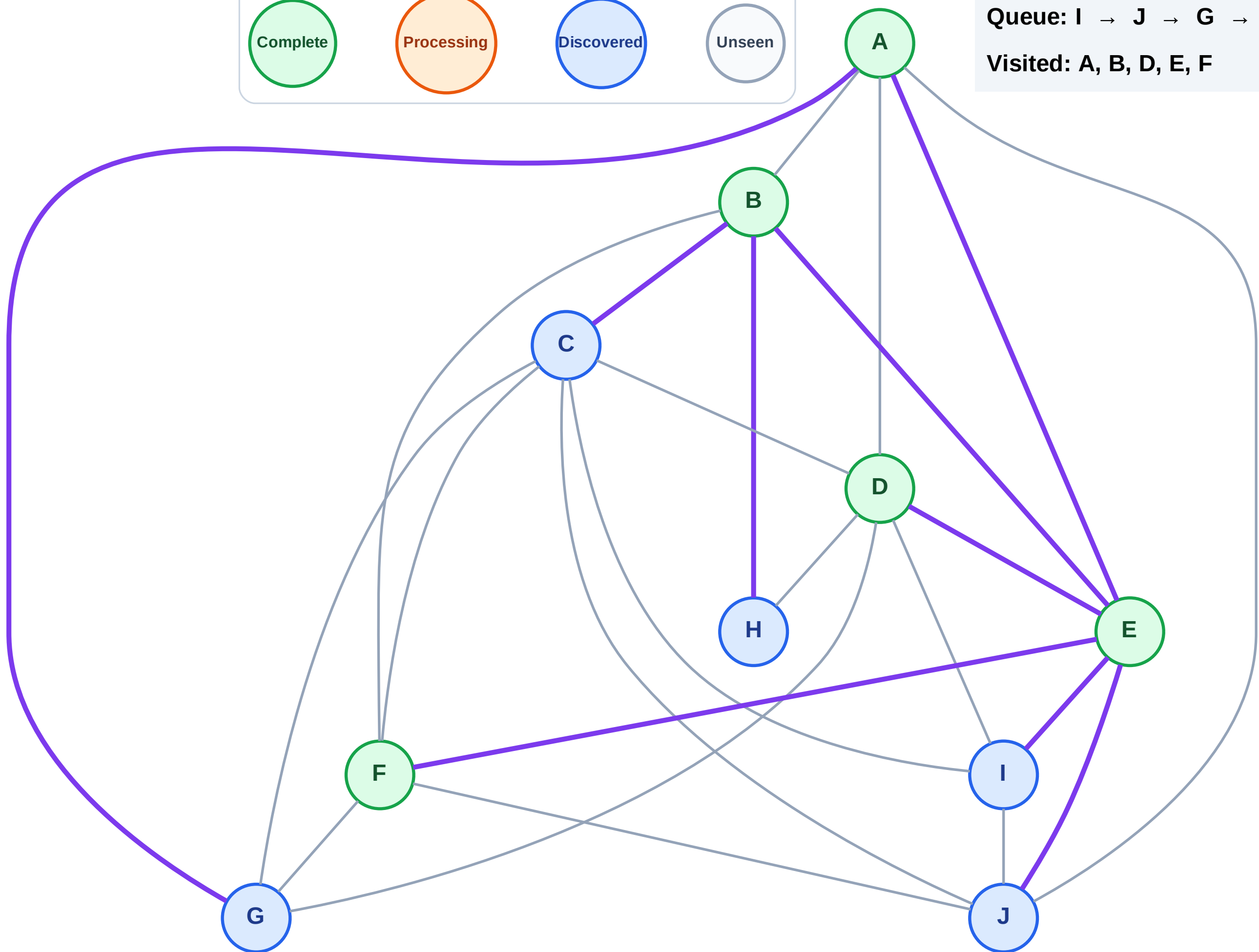
Step 11: No new neighbors from F

Legend



Queue: I → J → G → C → H

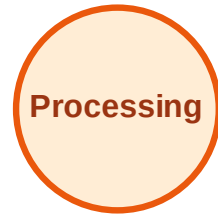
Visited: A, B, D, E, F



BFS Traversal

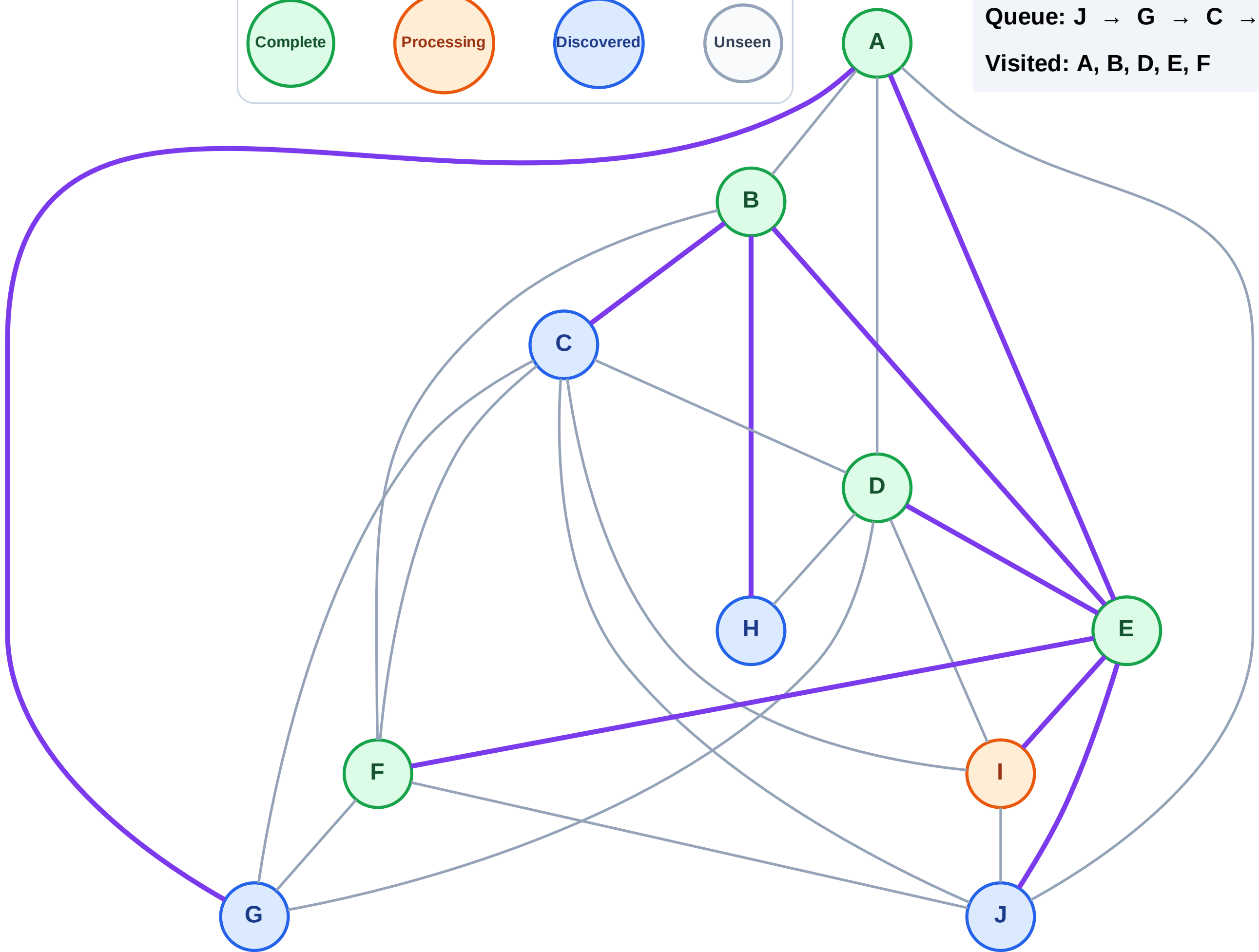
Step 12: Process node I

Legend



Queue: J → G → C → H

Visited: A, B, D, E, F



BFS Traversal

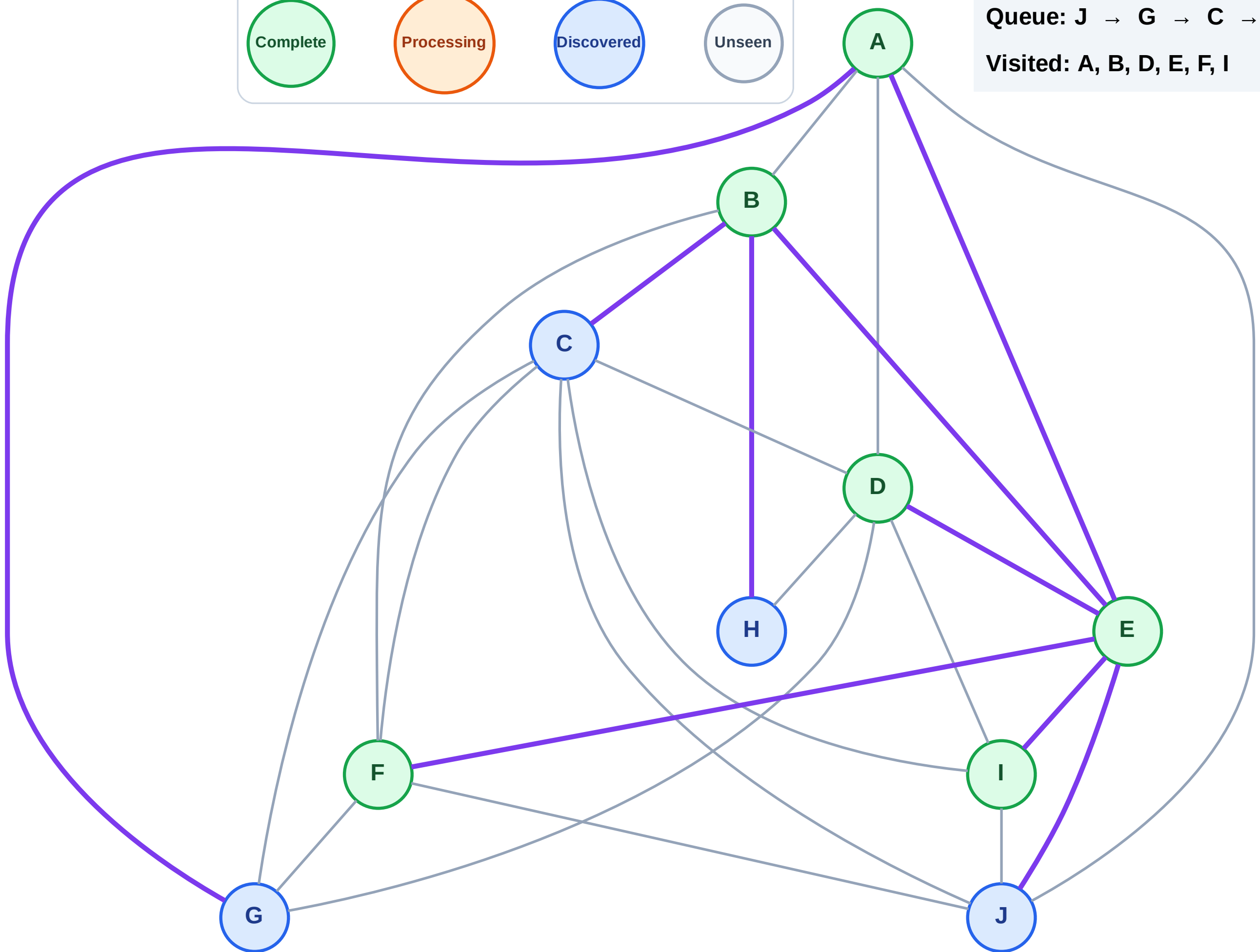
Step 13: No new neighbors from I

Legend



Queue: J → G → C → H

Visited: A, B, D, E, F, I



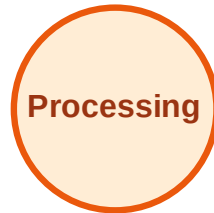
BFS Traversal

Step 14: Process node J

Legend



Complete



Processing

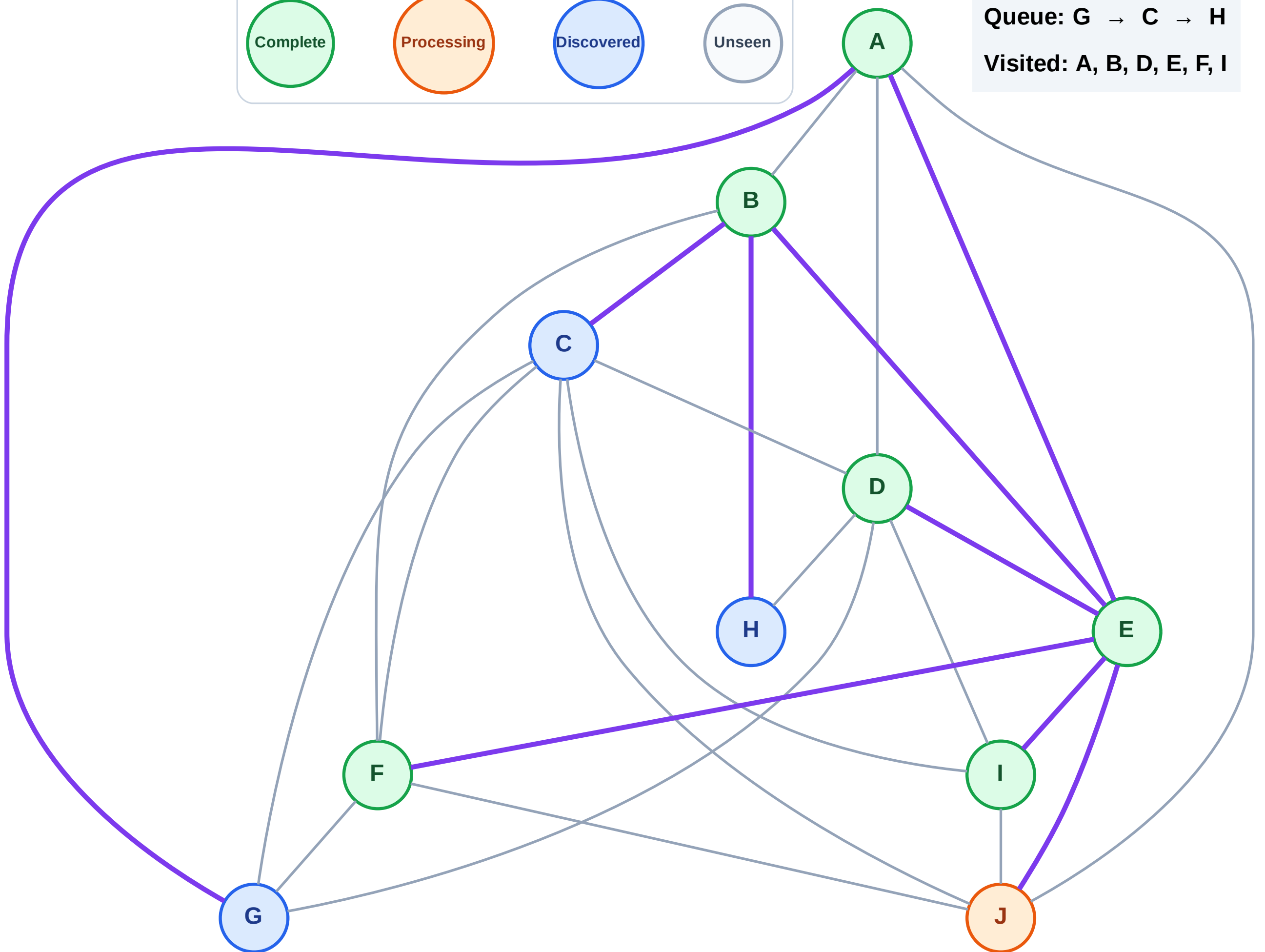


Discovered



Unseen

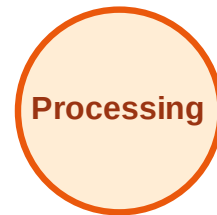
Queue: G → C → H
Visited: A, B, D, E, F, I



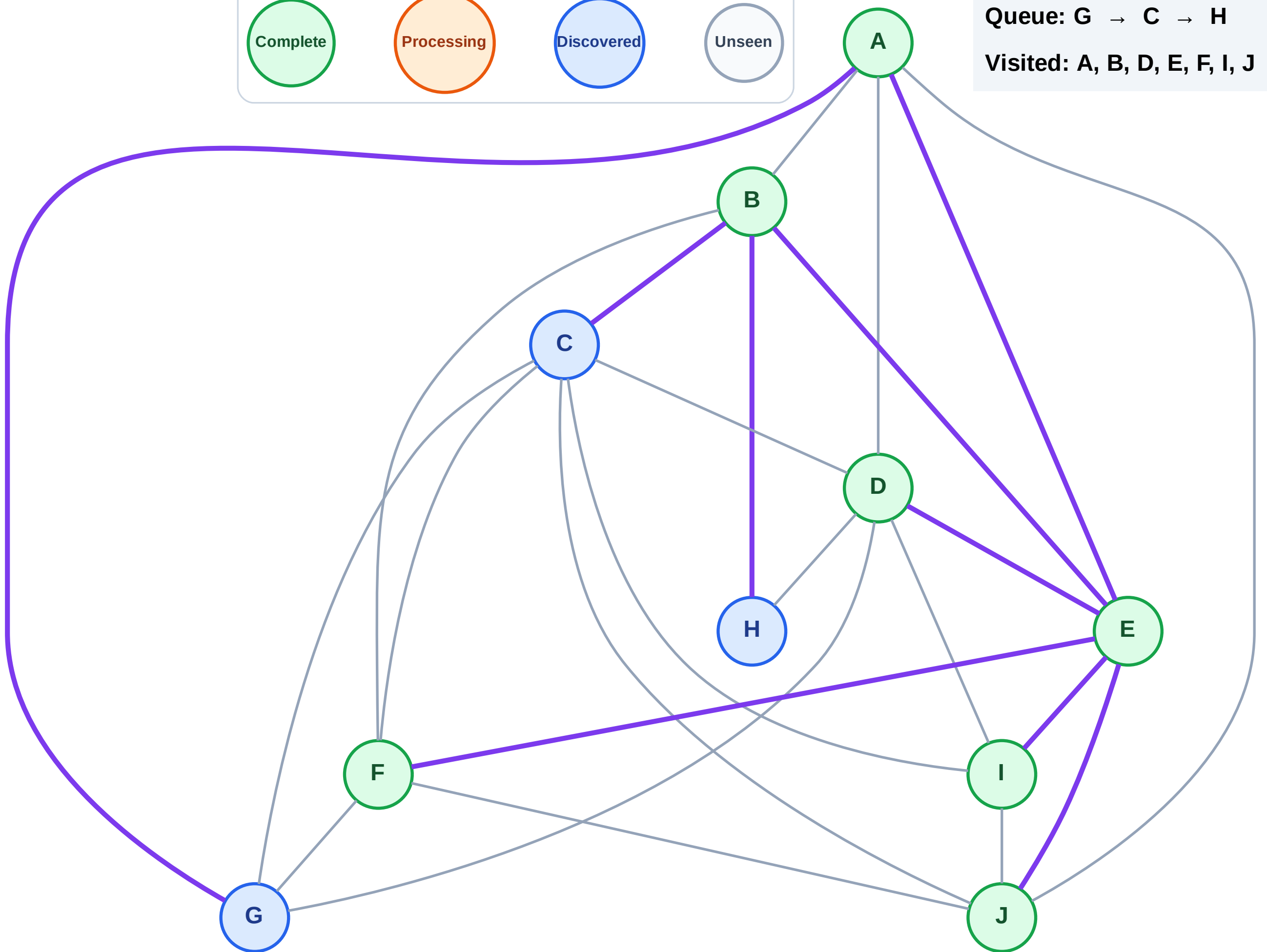
BFS Traversal

Step 15: No new neighbors from J

Legend



Queue: G → C → H
Visited: A, B, D, E, F, I, J



Step 16: Process node G

and

Legend

Complete

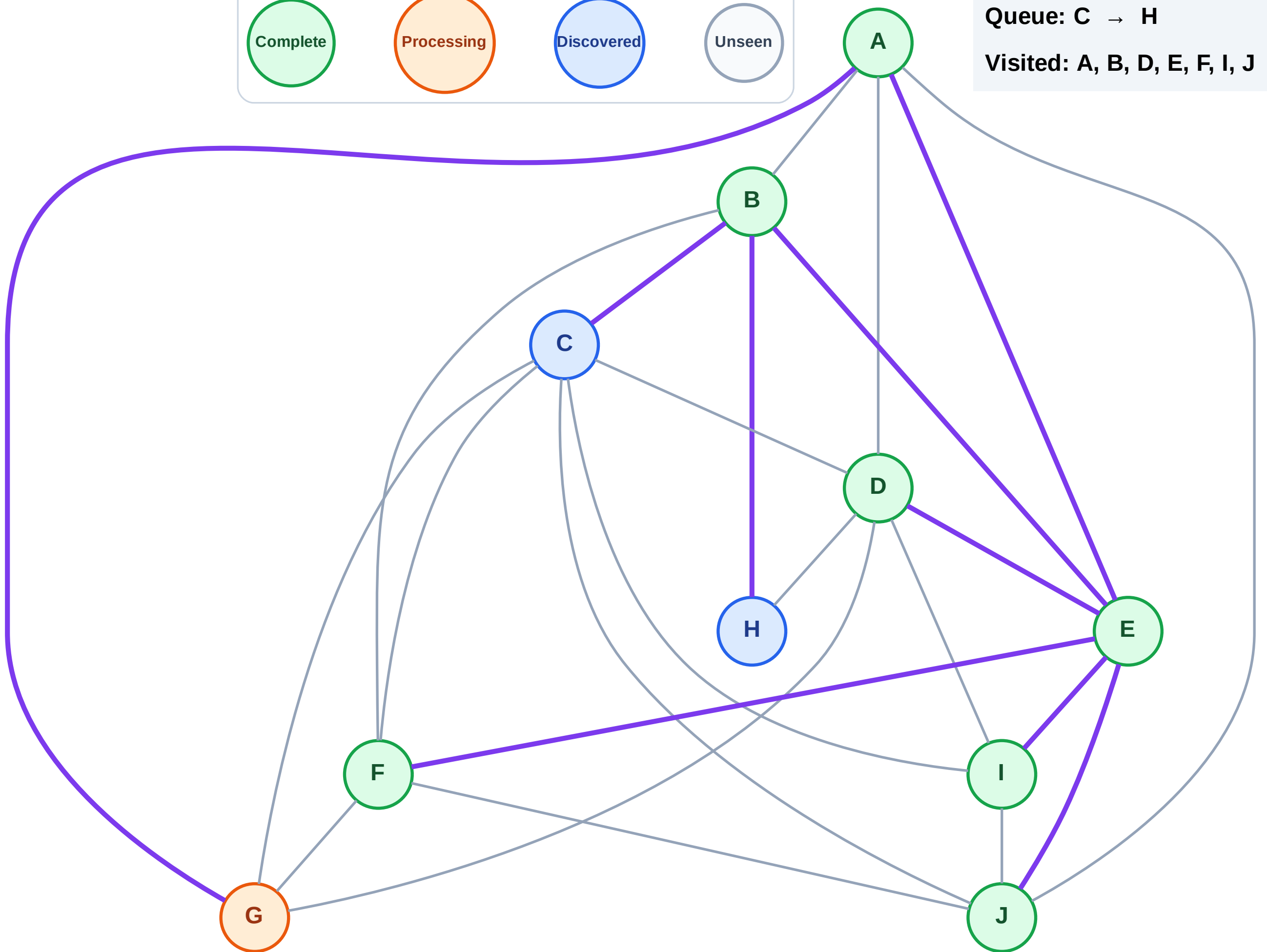
Processing

Discovered

Unseen

Queue: C → H

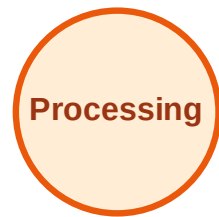
Visited: A, B, D, E, F, I, J



BFS Traversal

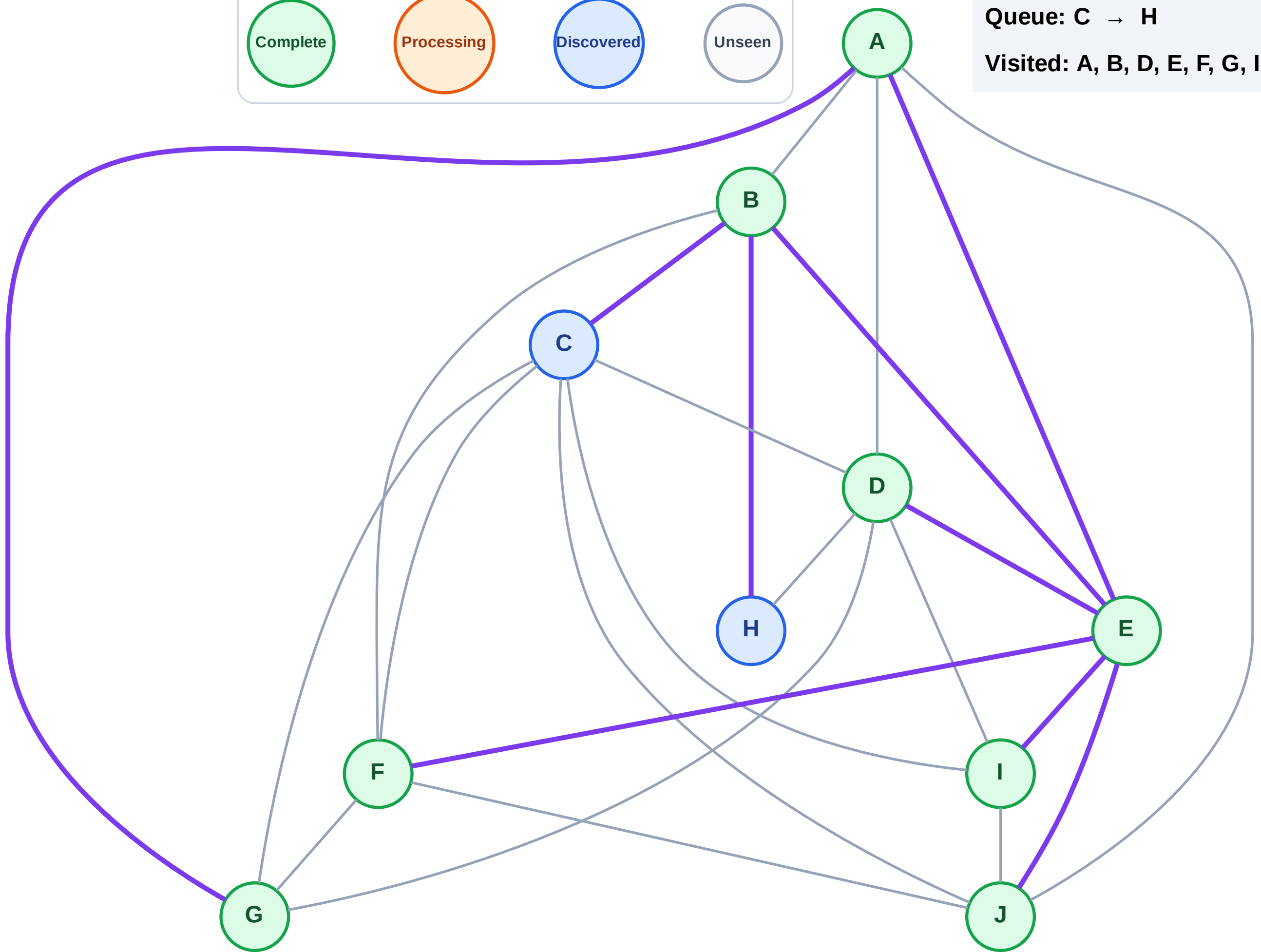
Step 17: No new neighbors from G

Legend



Queue: C → H

Visited: A, B, D, E, F, G, I, J

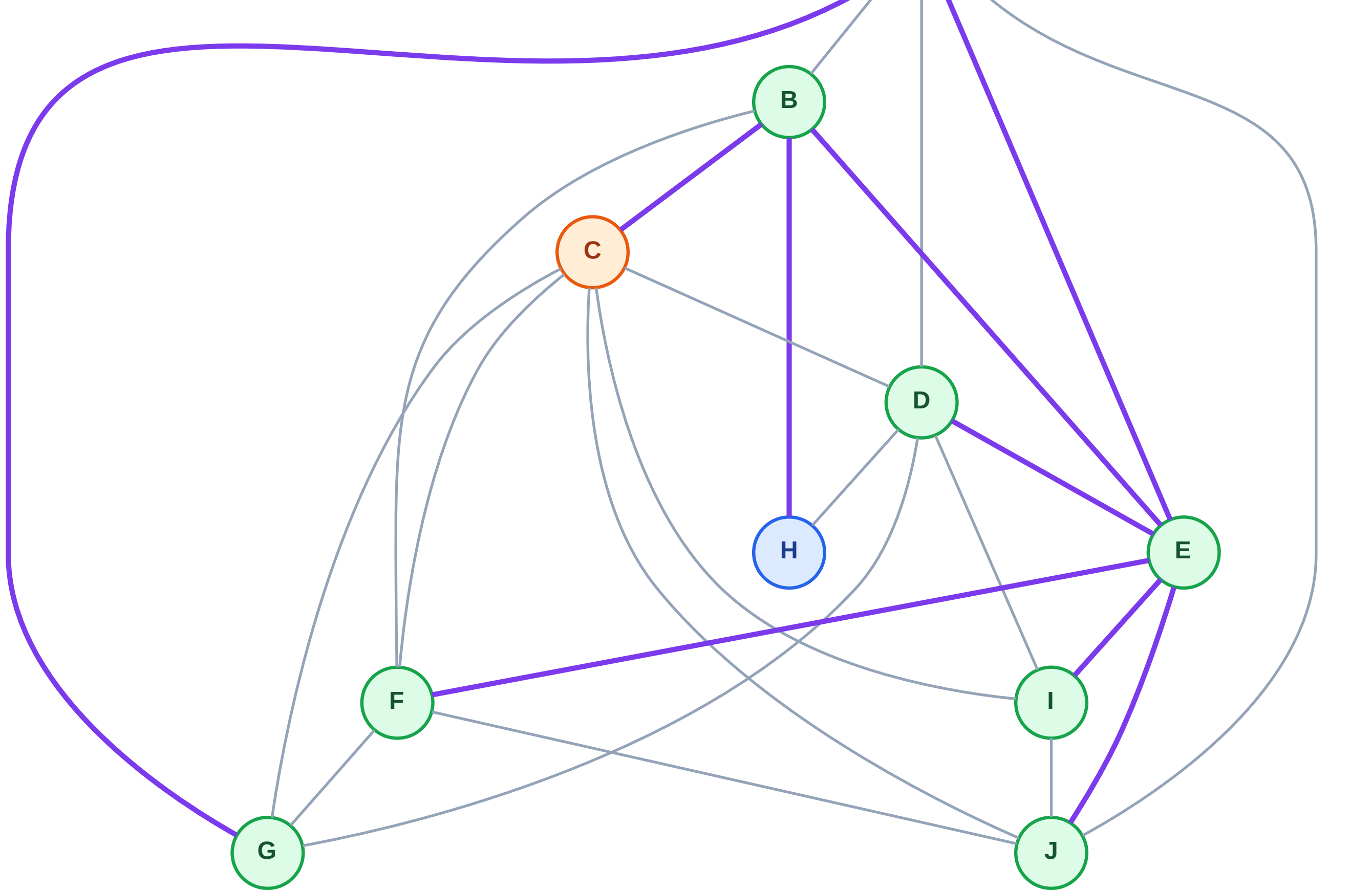


Step 18: Process node C

ed



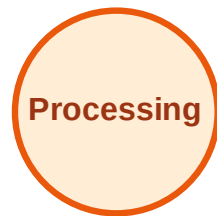
Visual Aids



BFS Traversal

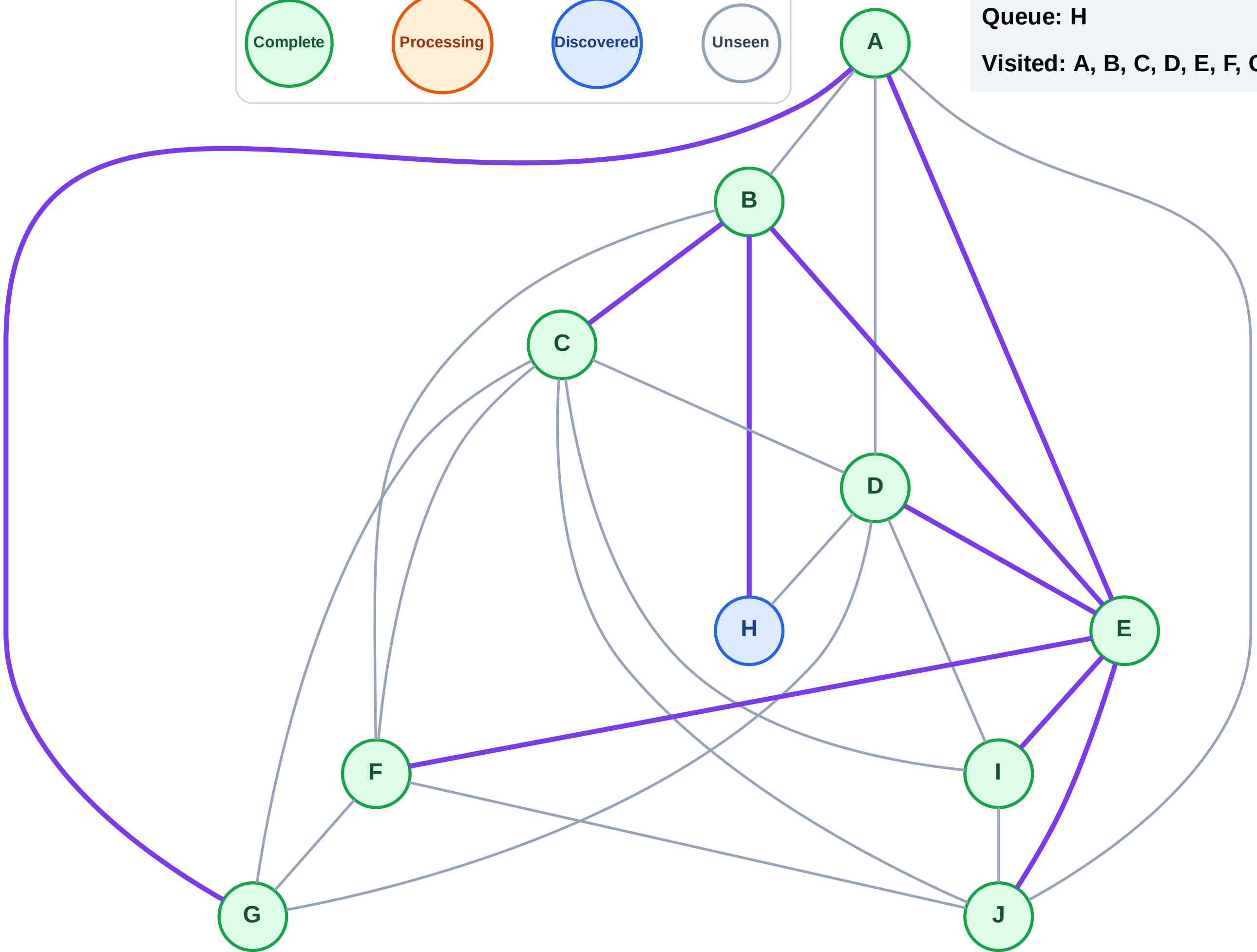
Step 19: No new neighbors from C

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



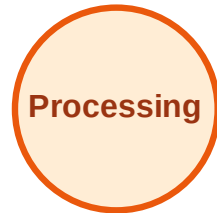
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



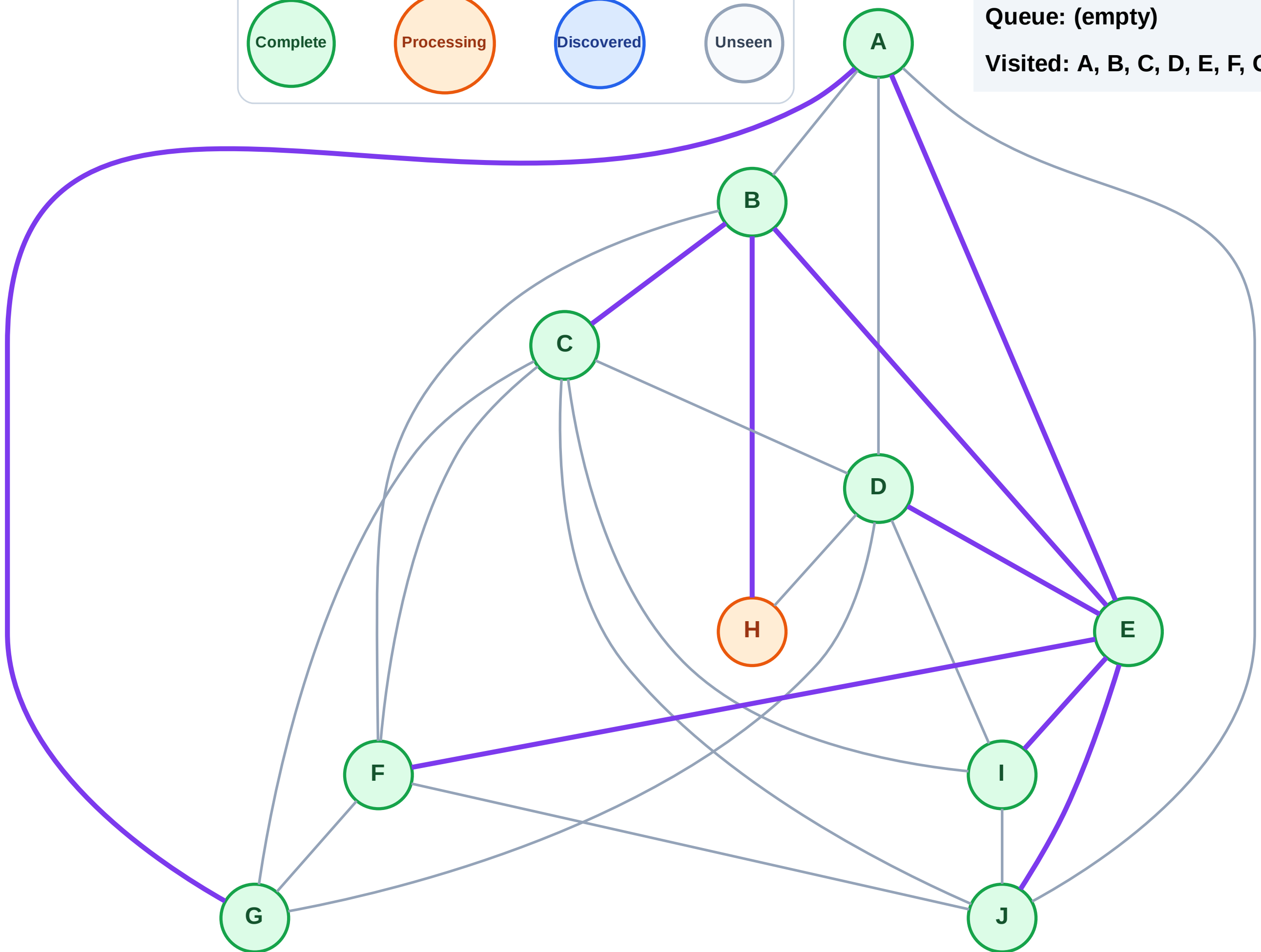
Discovered



Unseen

Queue: (empty)

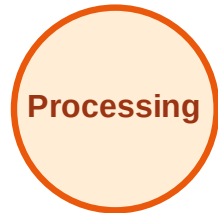
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

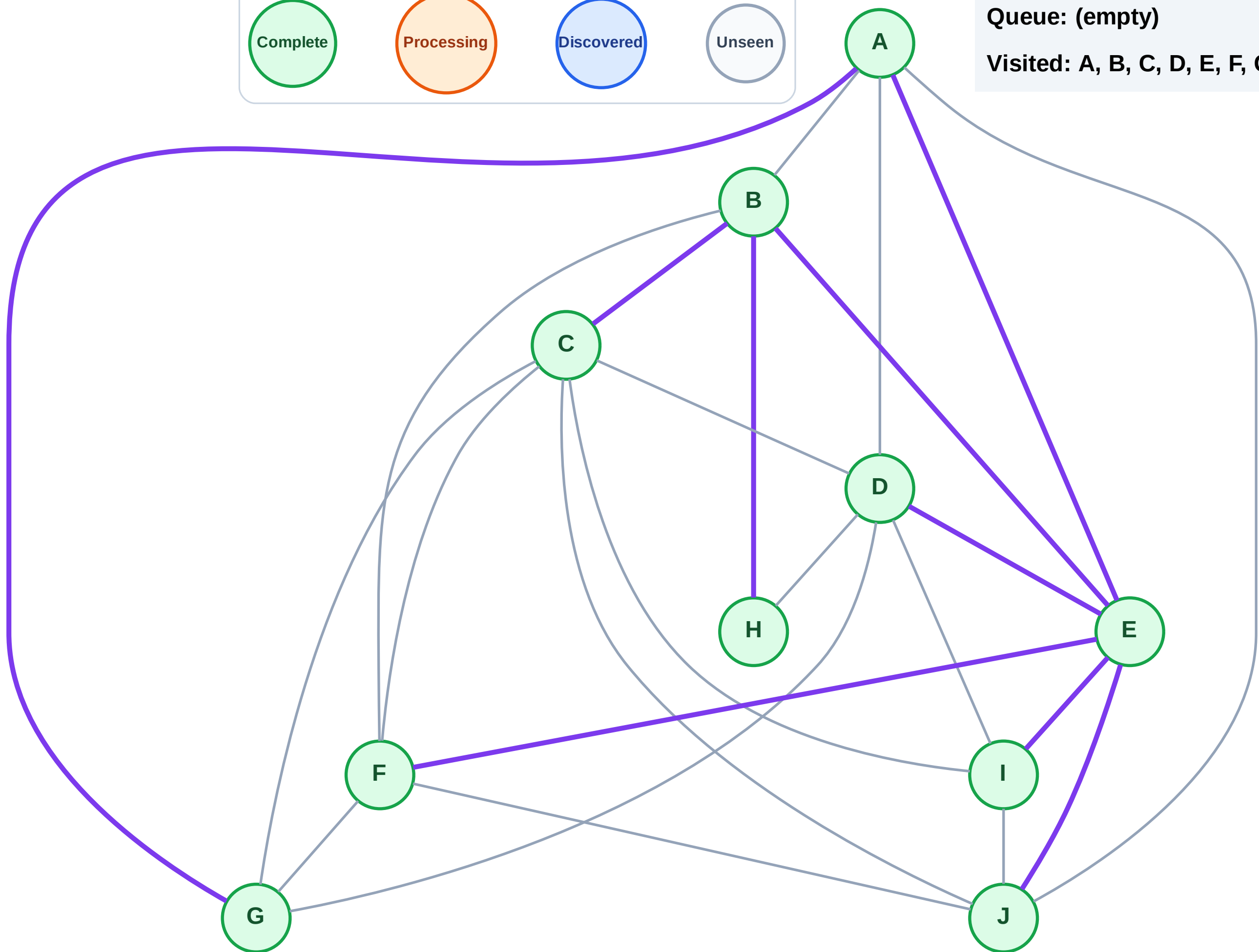
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



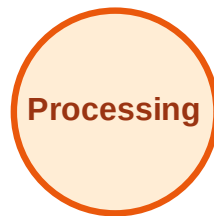
DFS Traversal

Step 1: Start at E

Legend



Complete



Processing



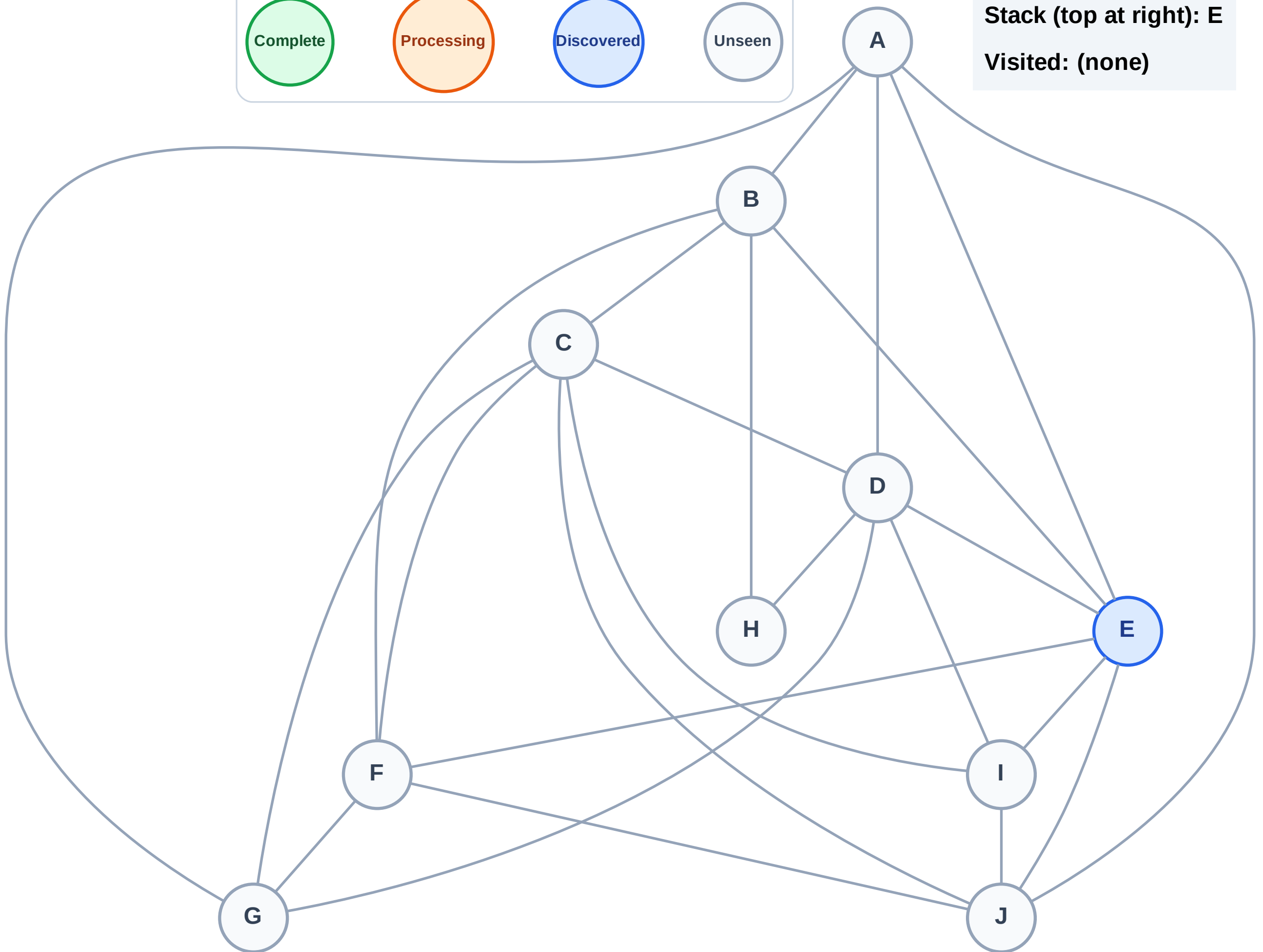
Discovered



Unseen

Stack (top at right): E

Visited: (none)



DFS Traversal

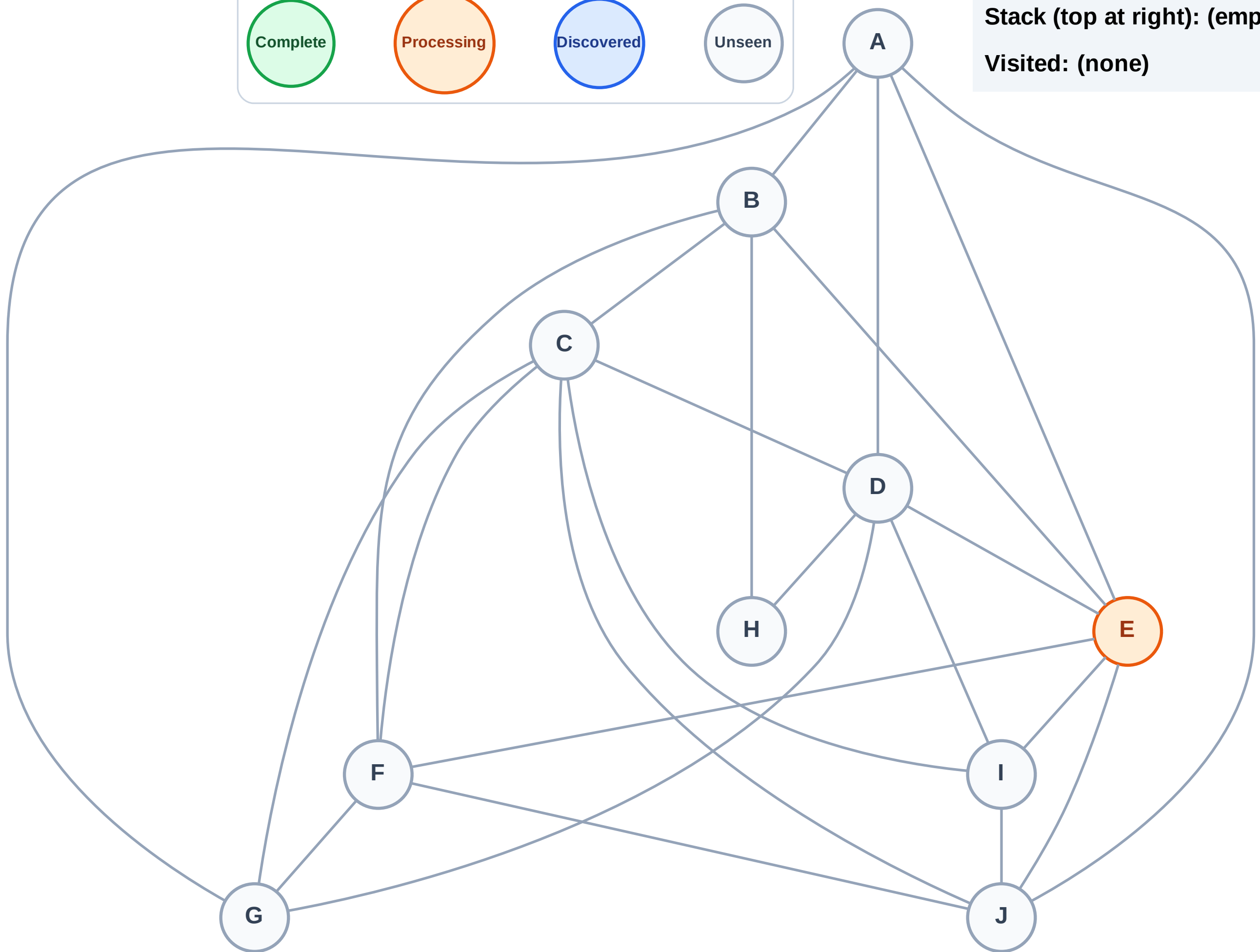
Step 2: Process node E

Legend



Stack (top at right): (empty)

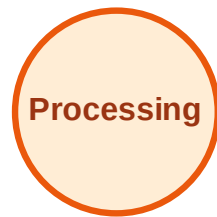
Visited: (none)



DFS Traversal

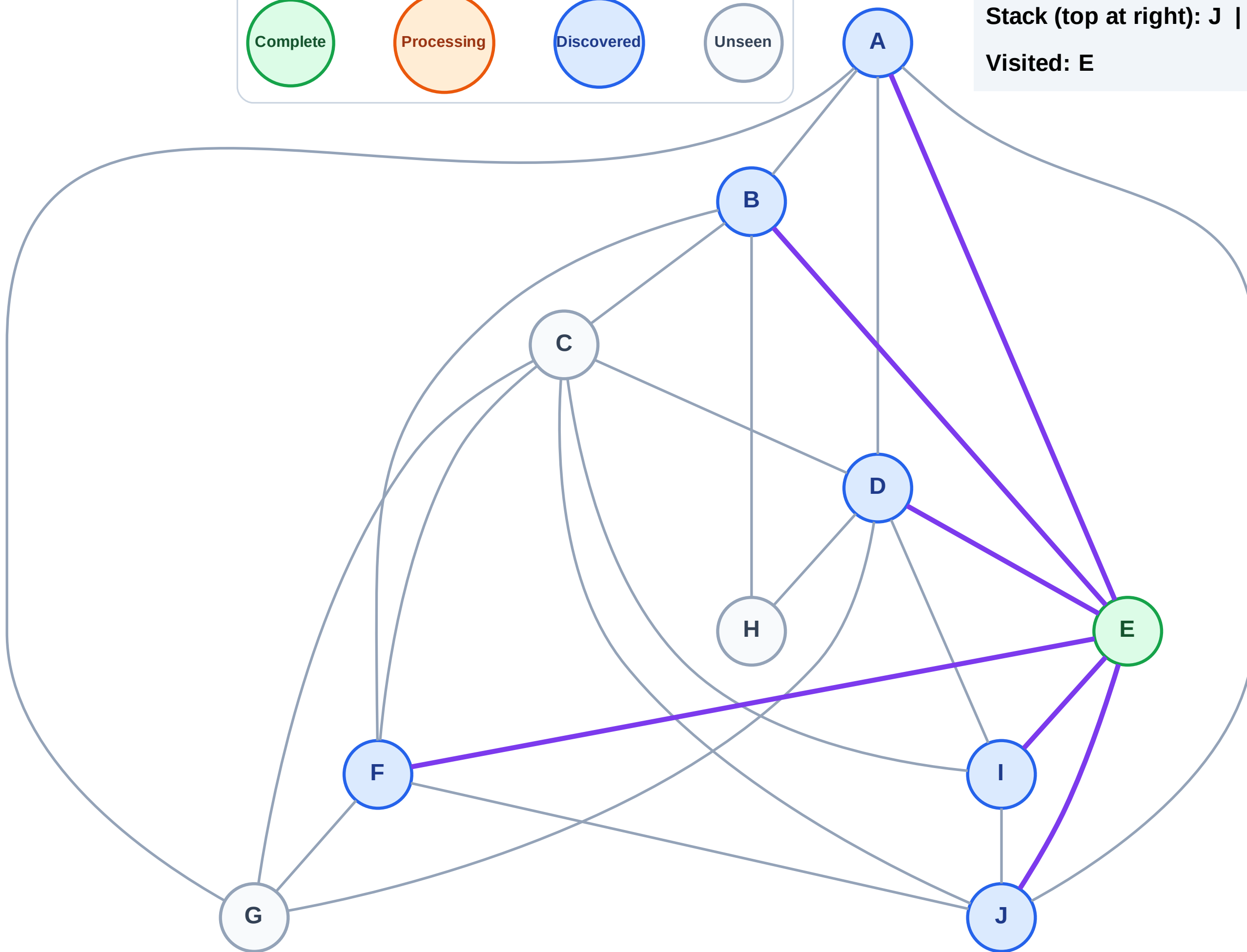
Step 3: Discover J, I, F, D, B, A from E

Legend



Stack (top at right): J | I | F | D | B | A

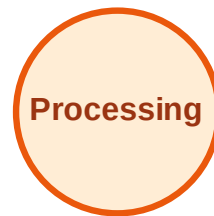
Visited: E



DFS Traversal

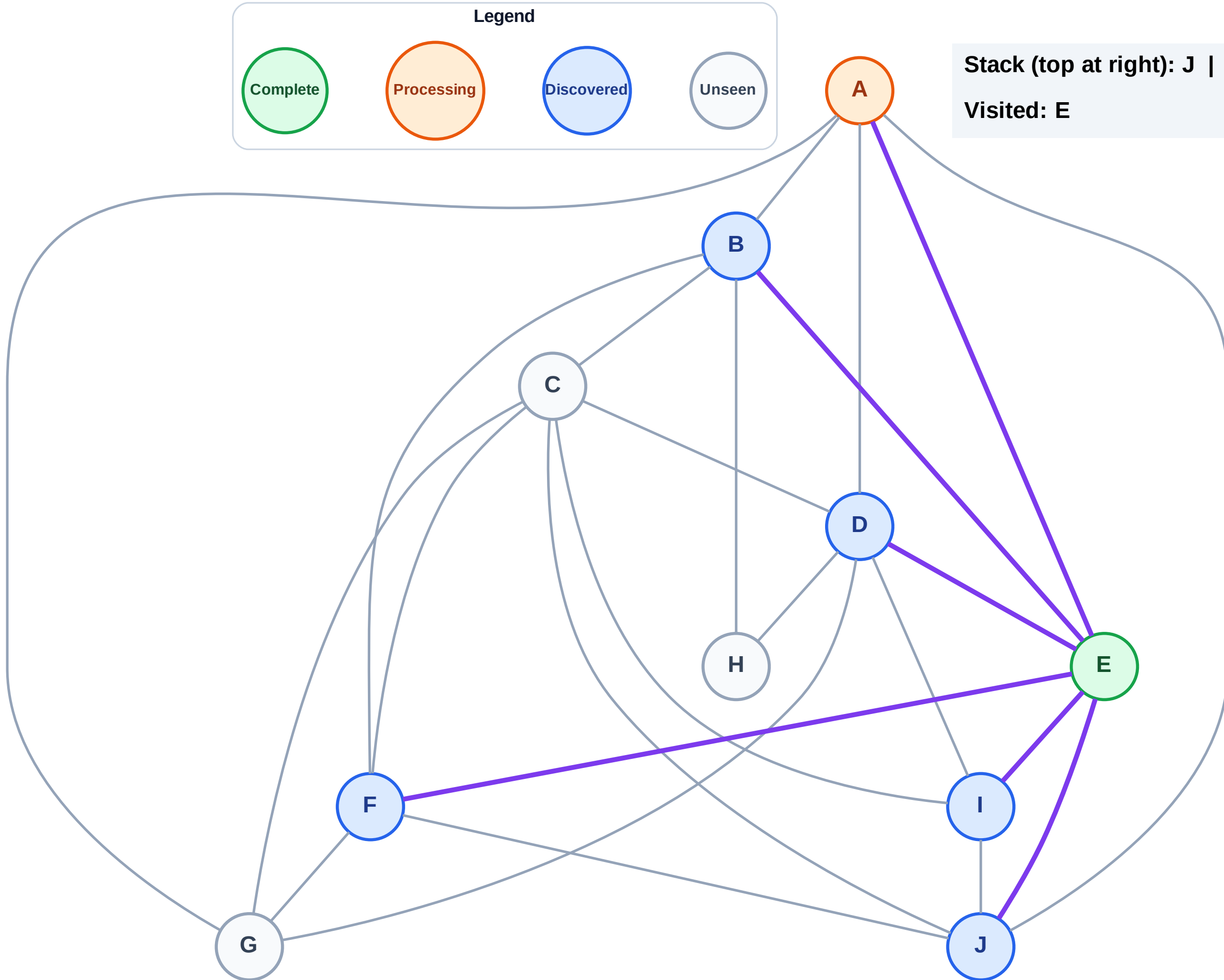
Step 4: Process node A

Legend



Stack (top at right): J | I | F | D | B

Visited: E



DFS Traversal

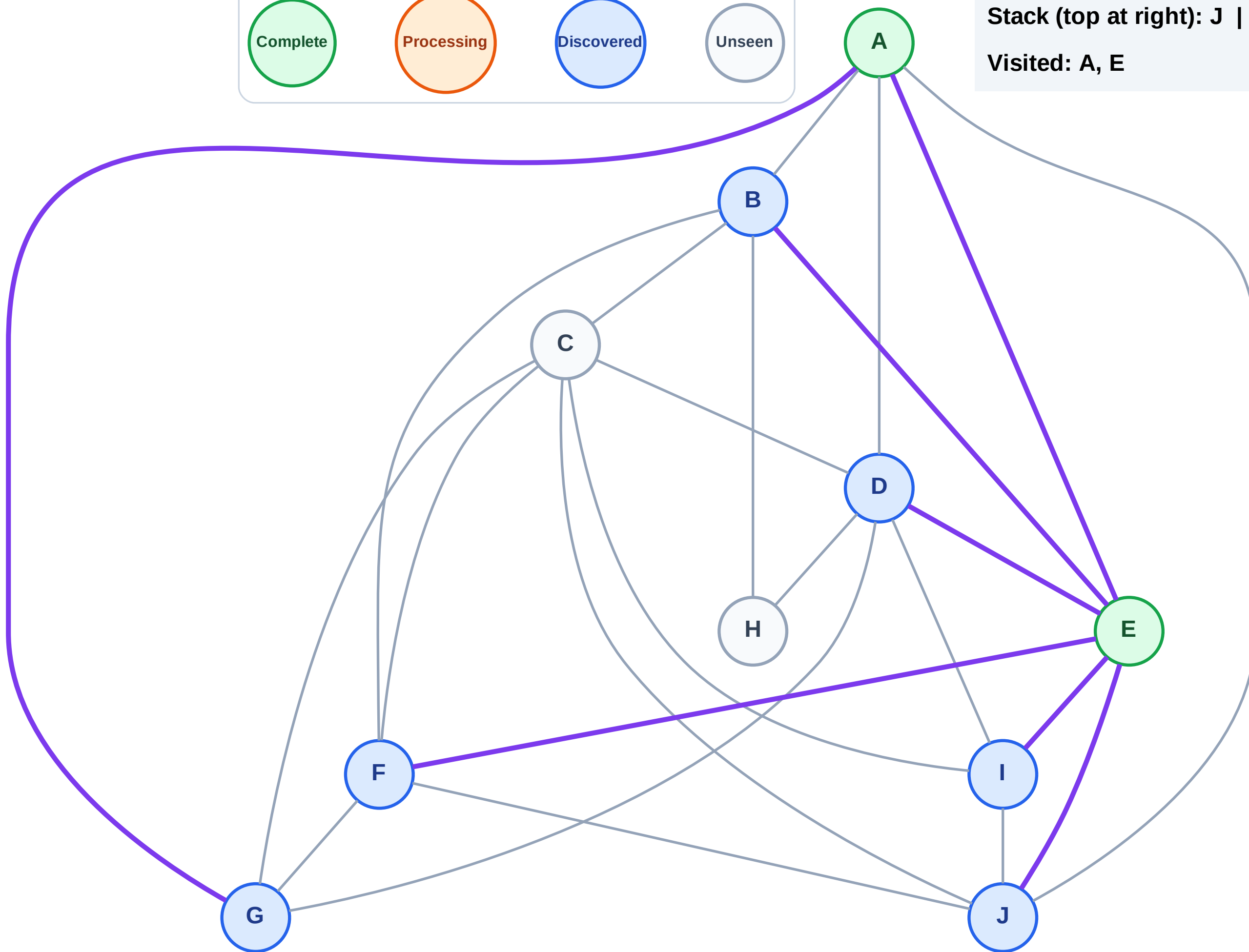
Step 5: Discover G from A

Legend



Stack (top at right): J | I | F | D | B | G

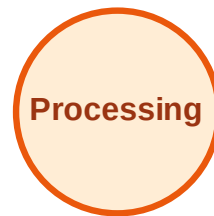
Visited: A, E



DFS Traversal

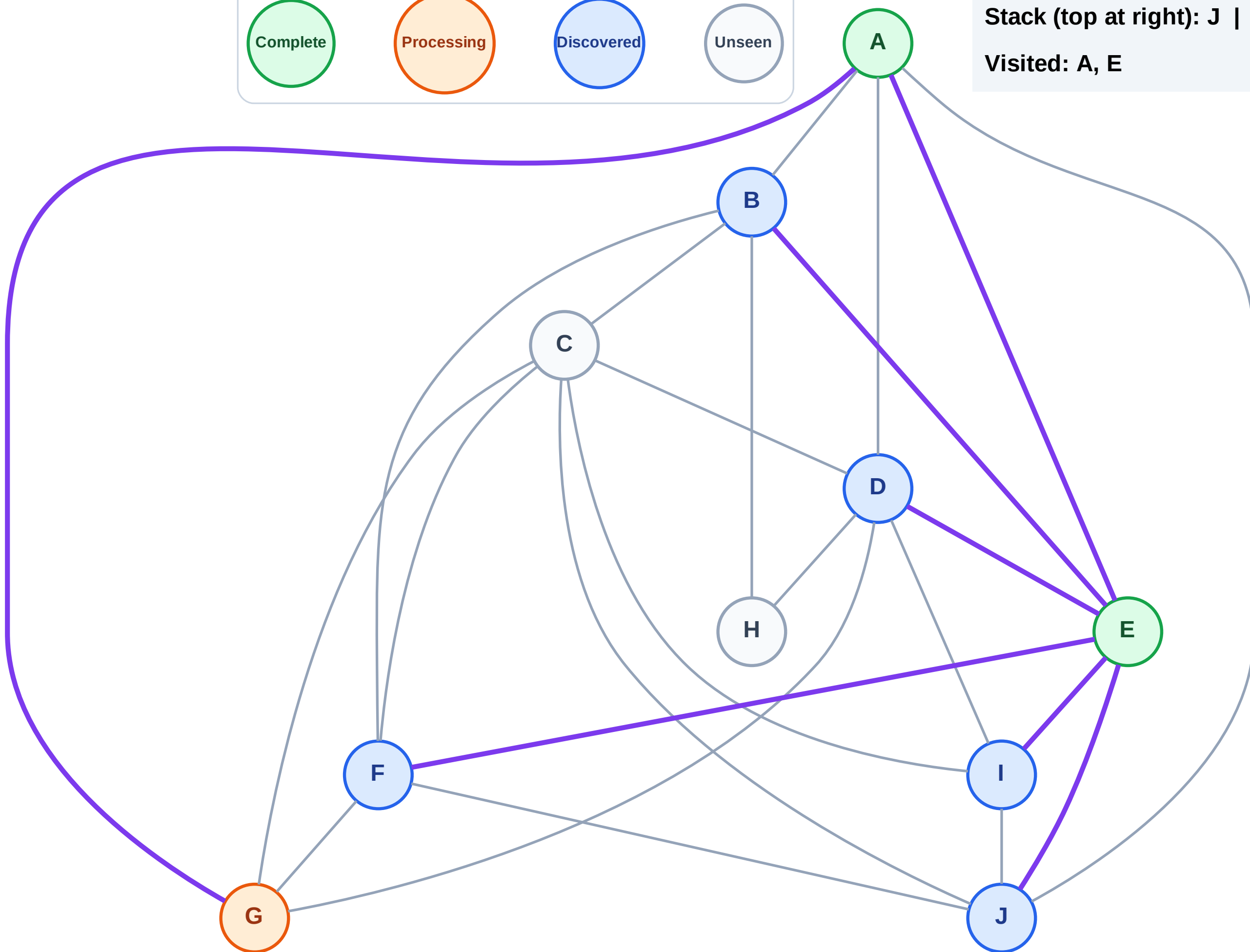
Step 6: Process node G

Legend



Stack (top at right): J | I | F | D | B

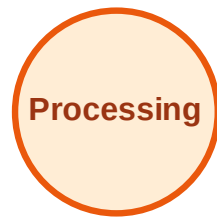
Visited: A, E



DFS Traversal

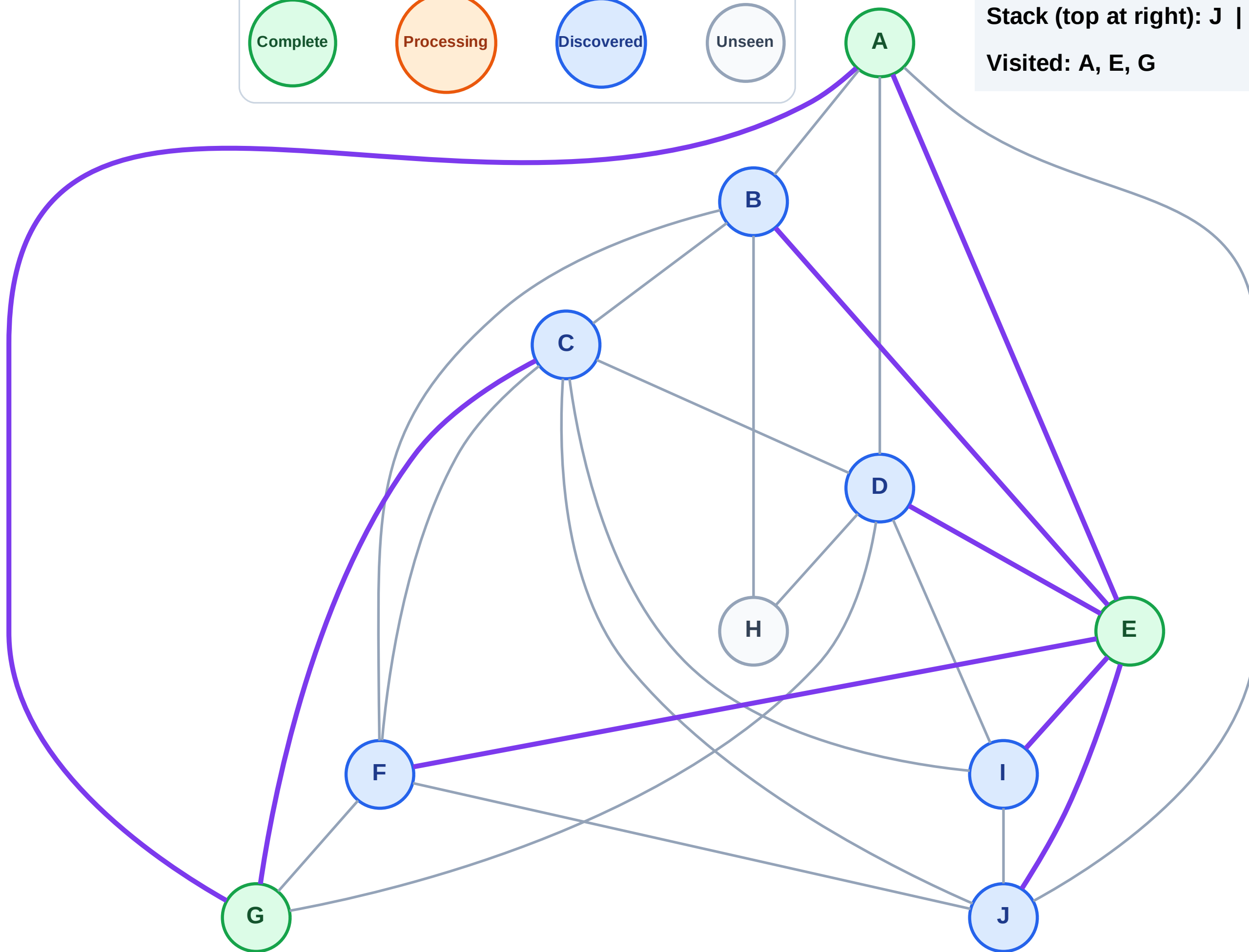
Step 7: Discover C from G

Legend



Stack (top at right): J | I | F | D | B | C

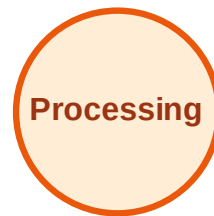
Visited: A, E, G



DFS Traversal

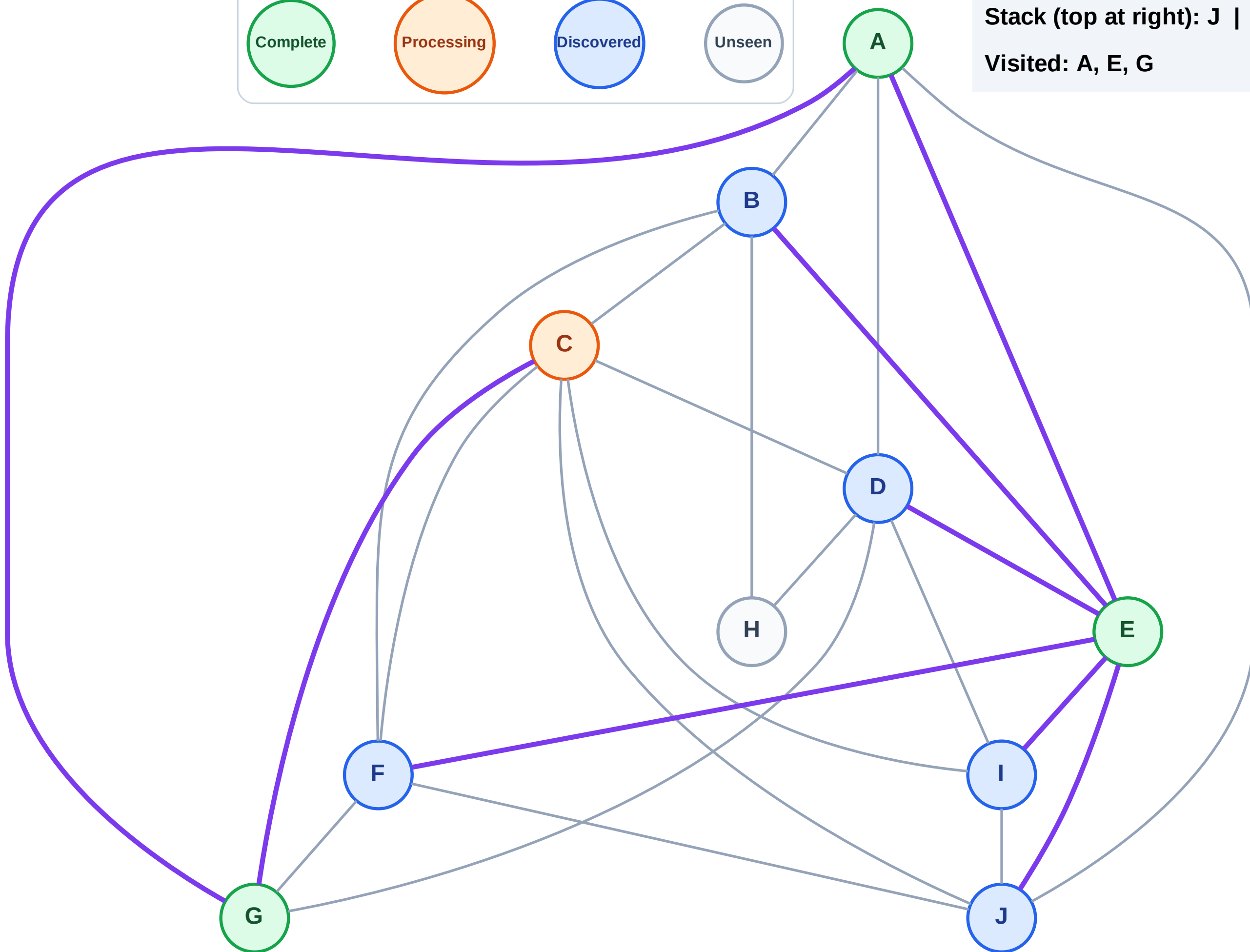
Step 8: Process node C

Legend



Stack (top at right): J | I | F | D | B

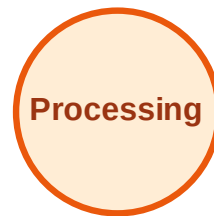
Visited: A, E, G



DFS Traversal

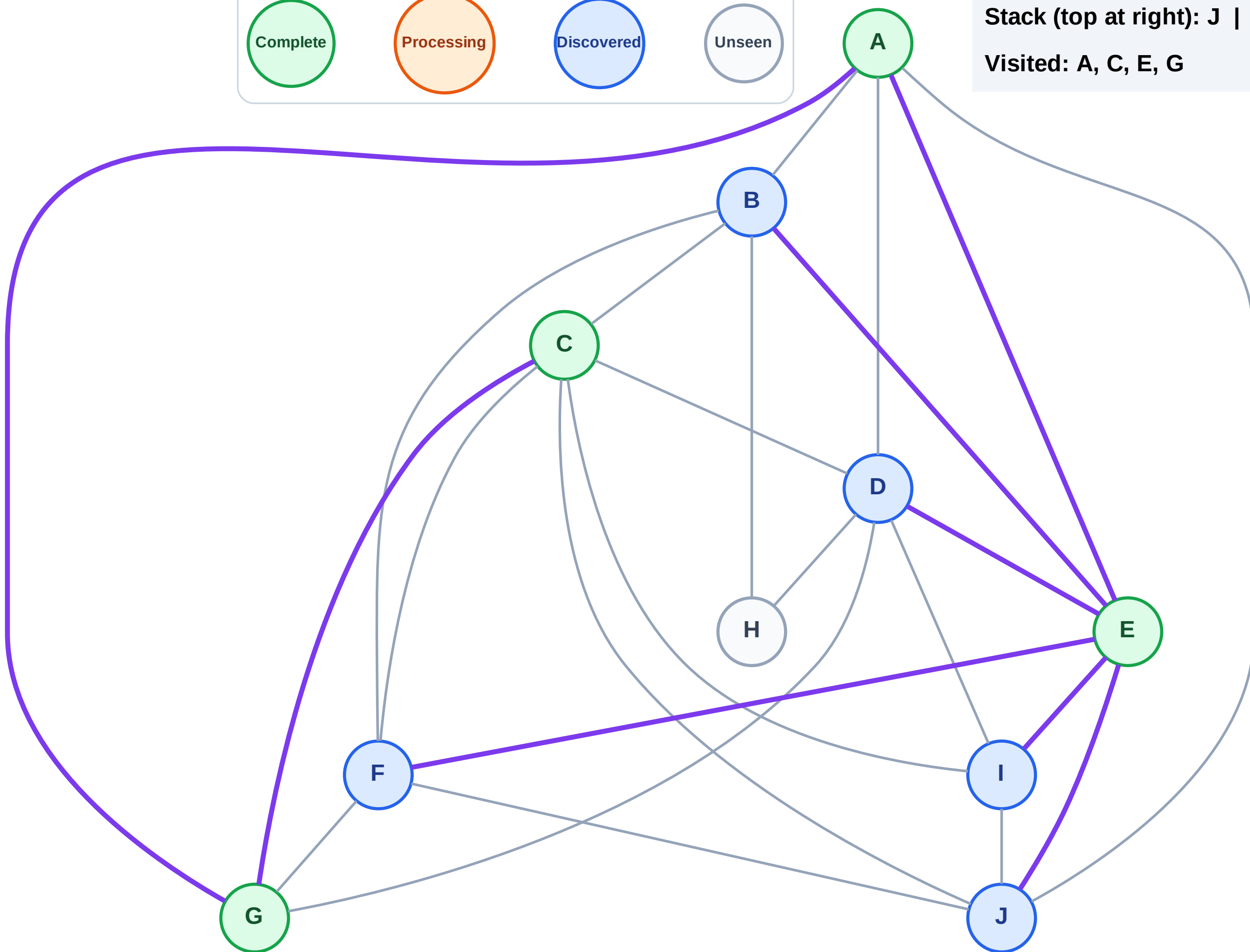
Step 9: No new neighbors from C

Legend



Stack (top at right): J | I | F | D | B

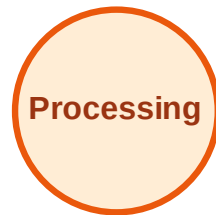
Visited: A, C, E, G



DFS Traversal

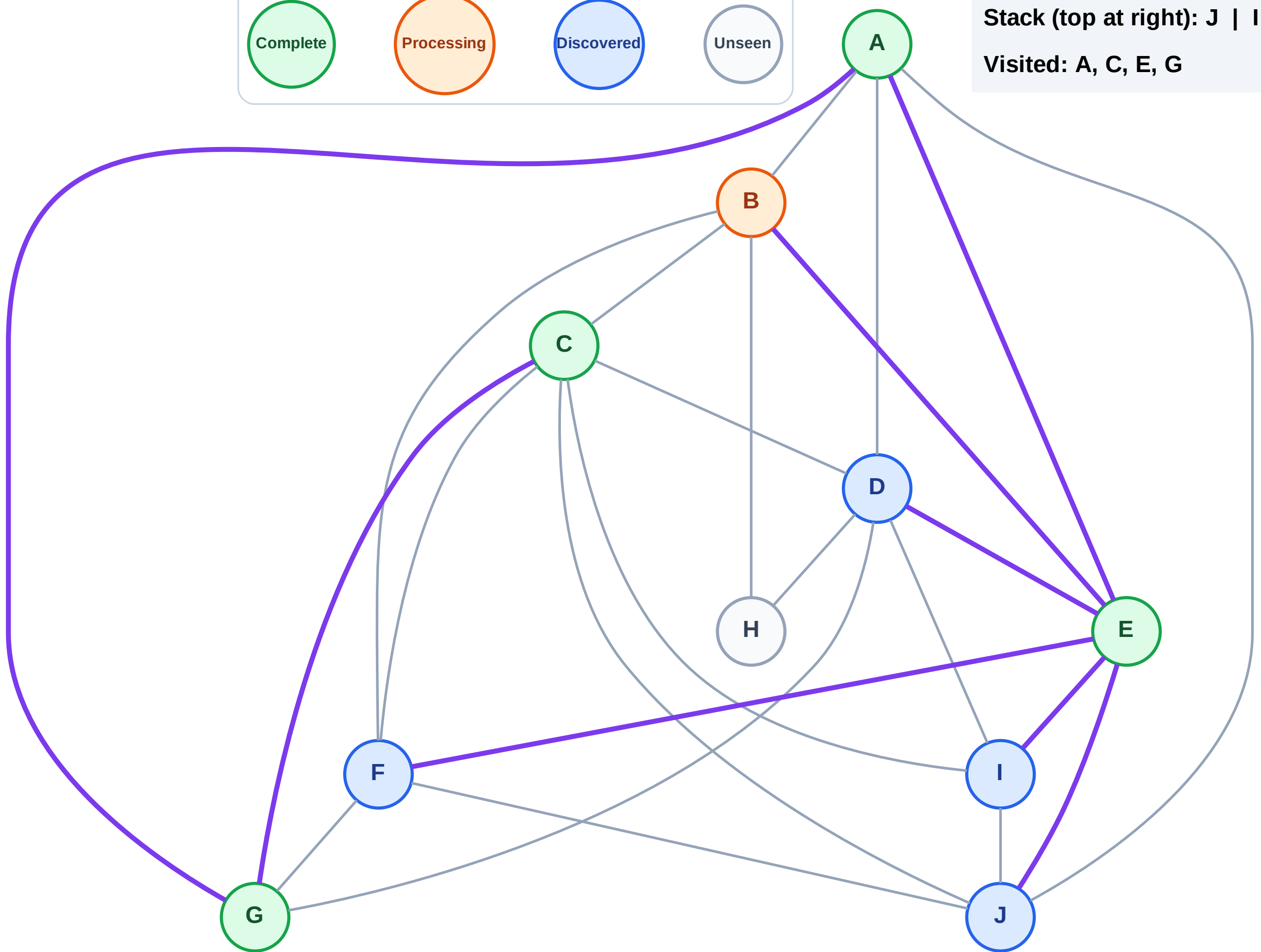
Step 10: Process node B

Legend



Stack (top at right): J | I | F | D

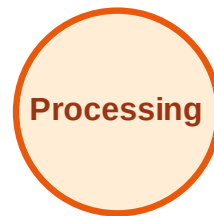
Visited: A, C, E, G



DFS Traversal

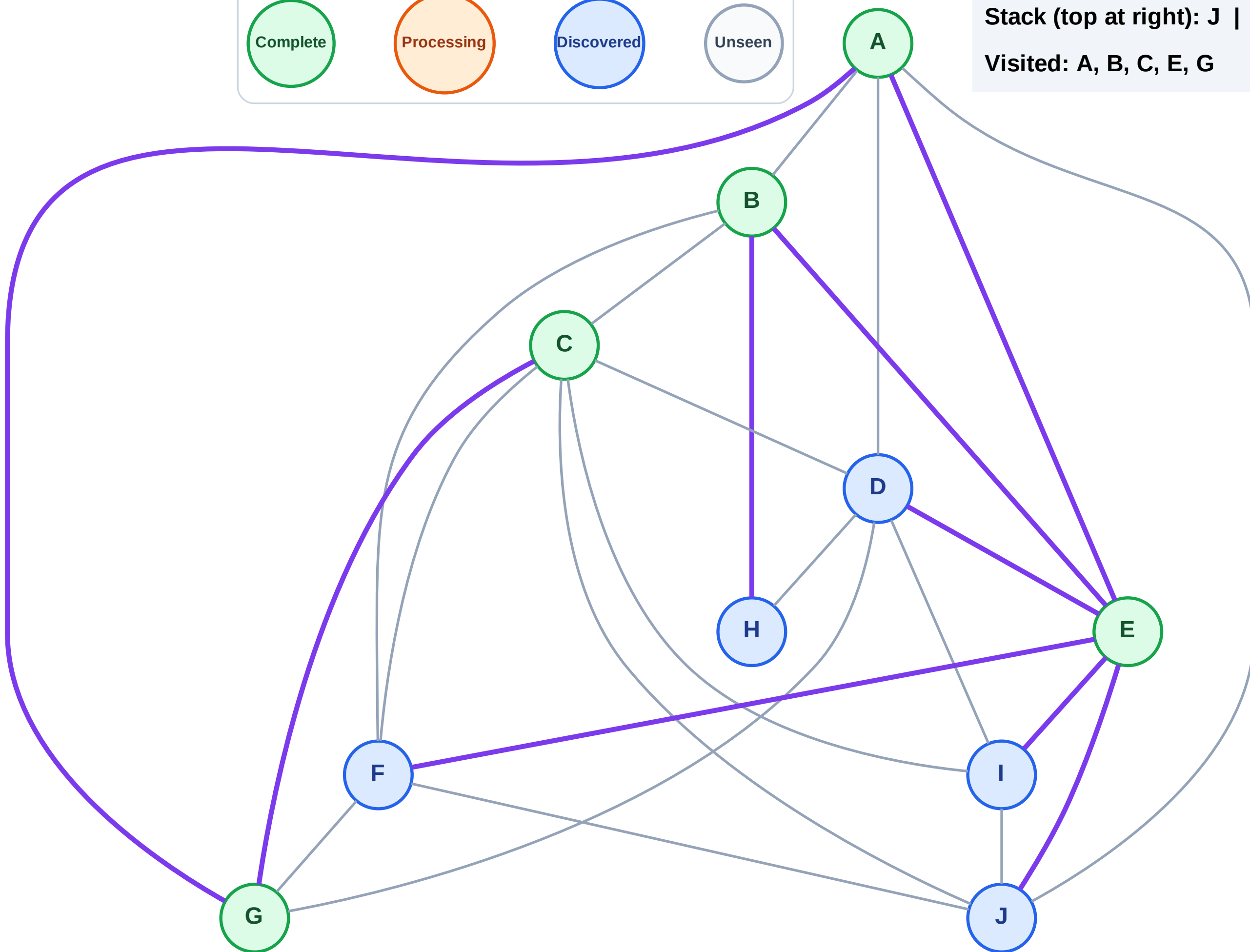
Step 11: Discover H from B

Legend



Stack (top at right): J | I | F | D | H

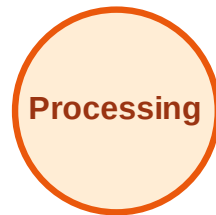
Visited: A, B, C, E, G



DFS Traversal

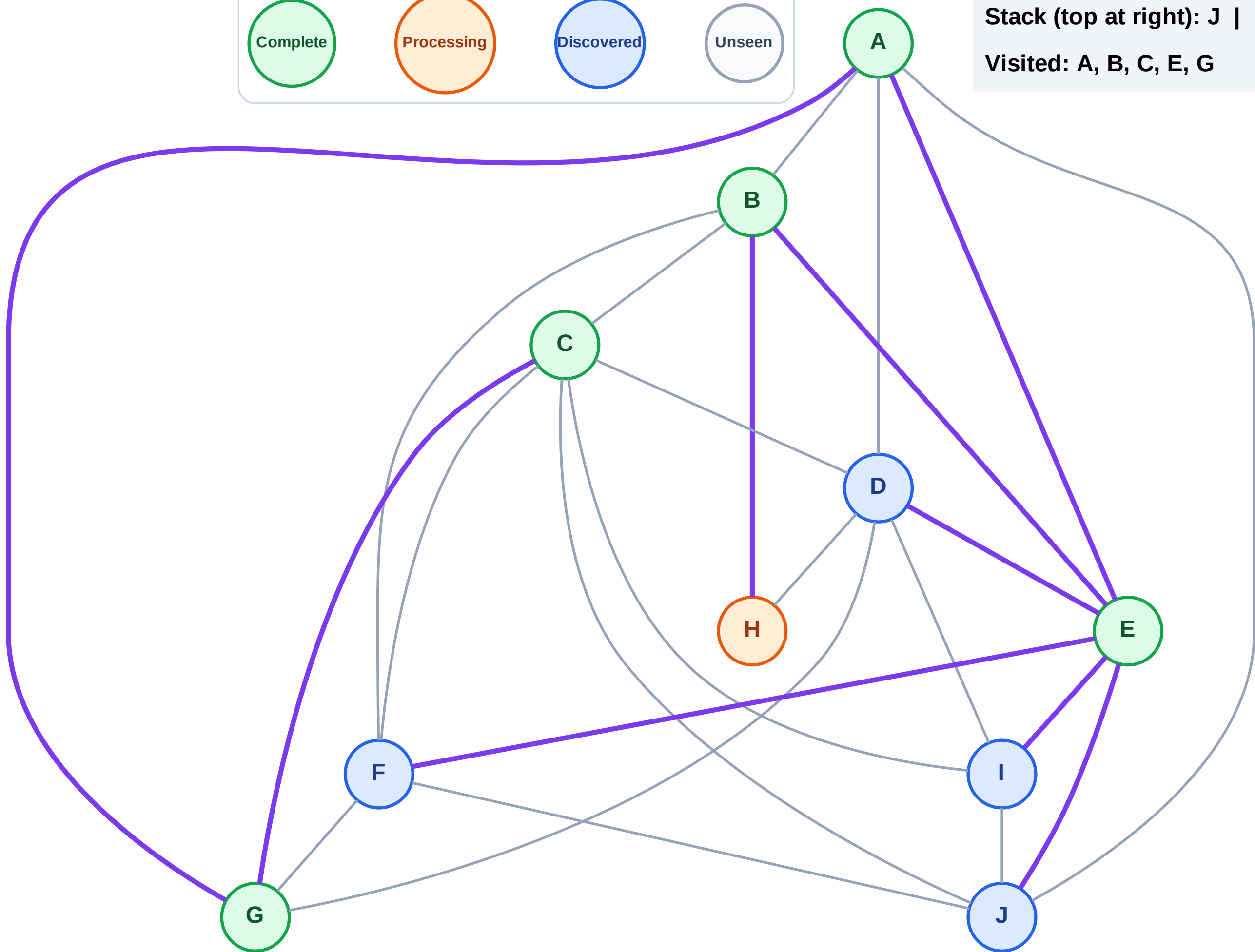
Step 12: Process node H

Legend



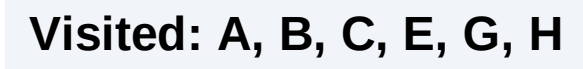
Stack (top at right): J | I | F | D

Visited: A, B, C, E, G



Step 13: No new neighbors from H

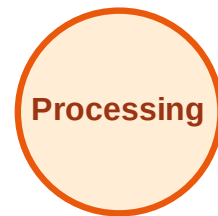
Legend



DFS Traversal

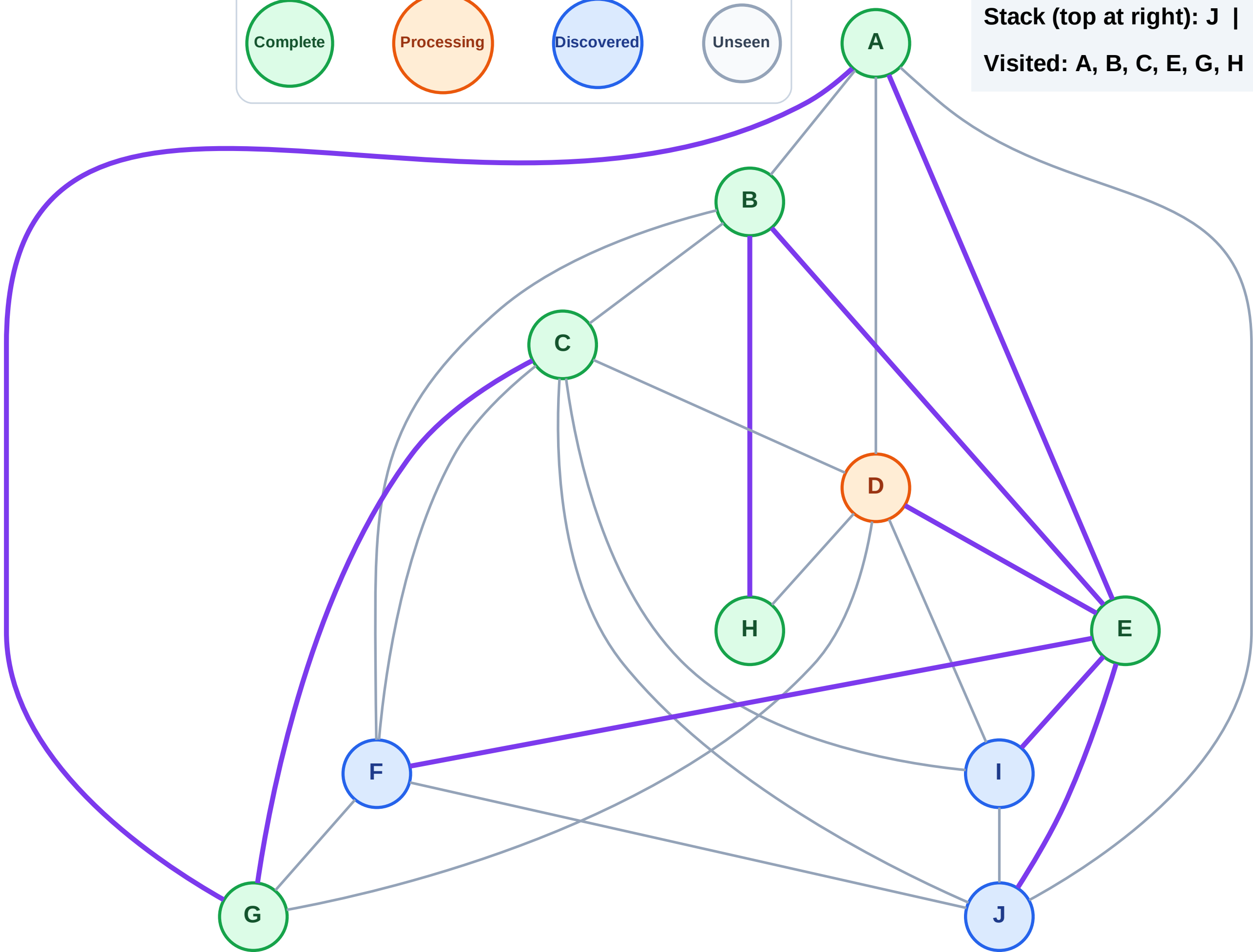
Step 14: Process node D

Legend



Stack (top at right): J | I | F

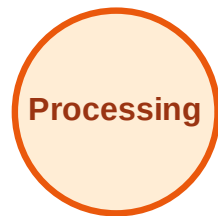
Visited: A, B, C, E, G, H



DFS Traversal

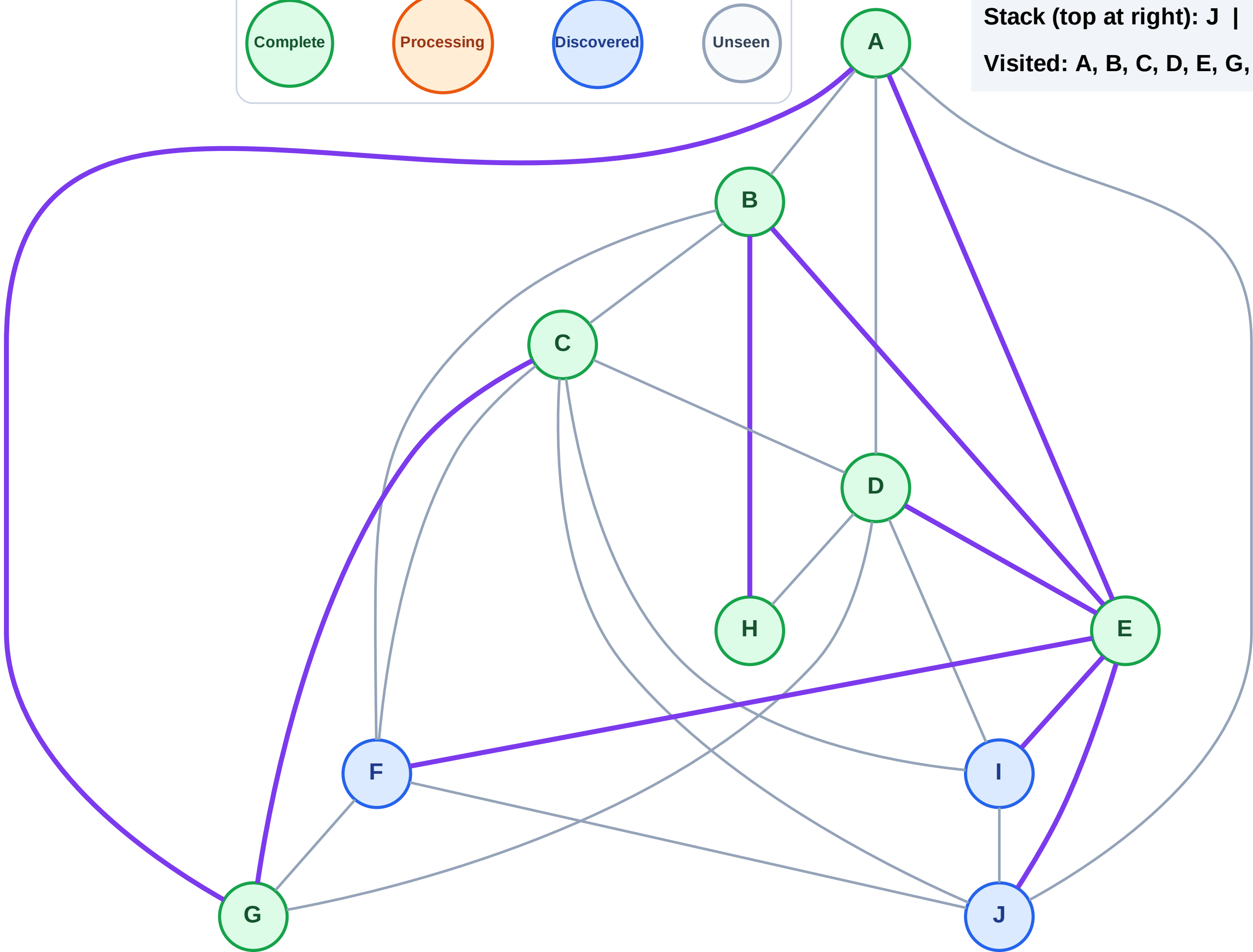
Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | F

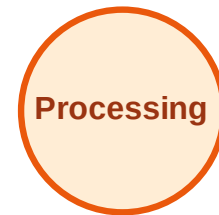
Visited: A, B, C, D, E, G, H



DFS Traversal

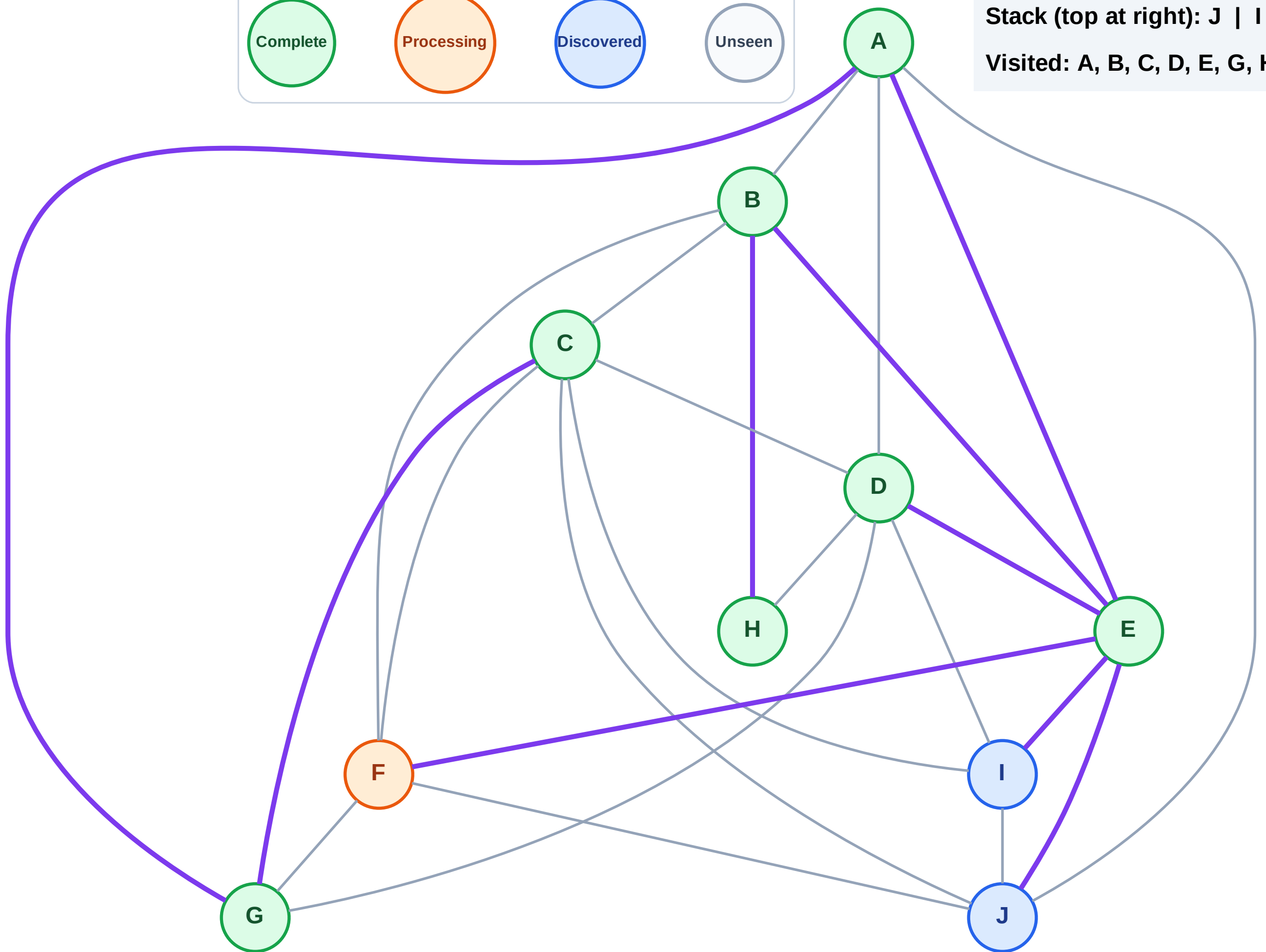
Step 16: Process node F

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, G, H



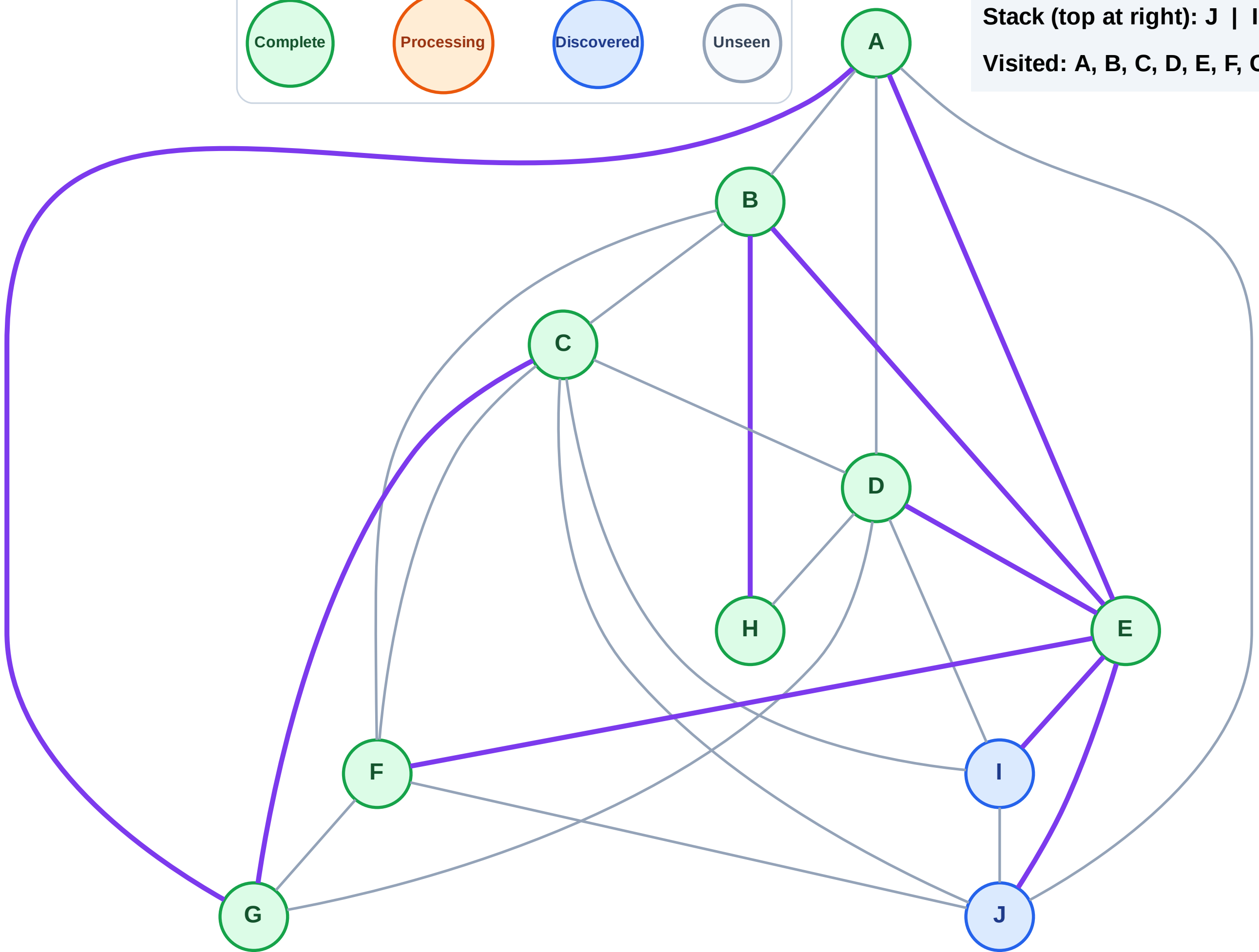
DFS Traversal

Step 17: No new neighbors from F

Legend



Stack (top at right): J | I
Visited: A, B, C, D, E, F, G, H



DFS Traversal

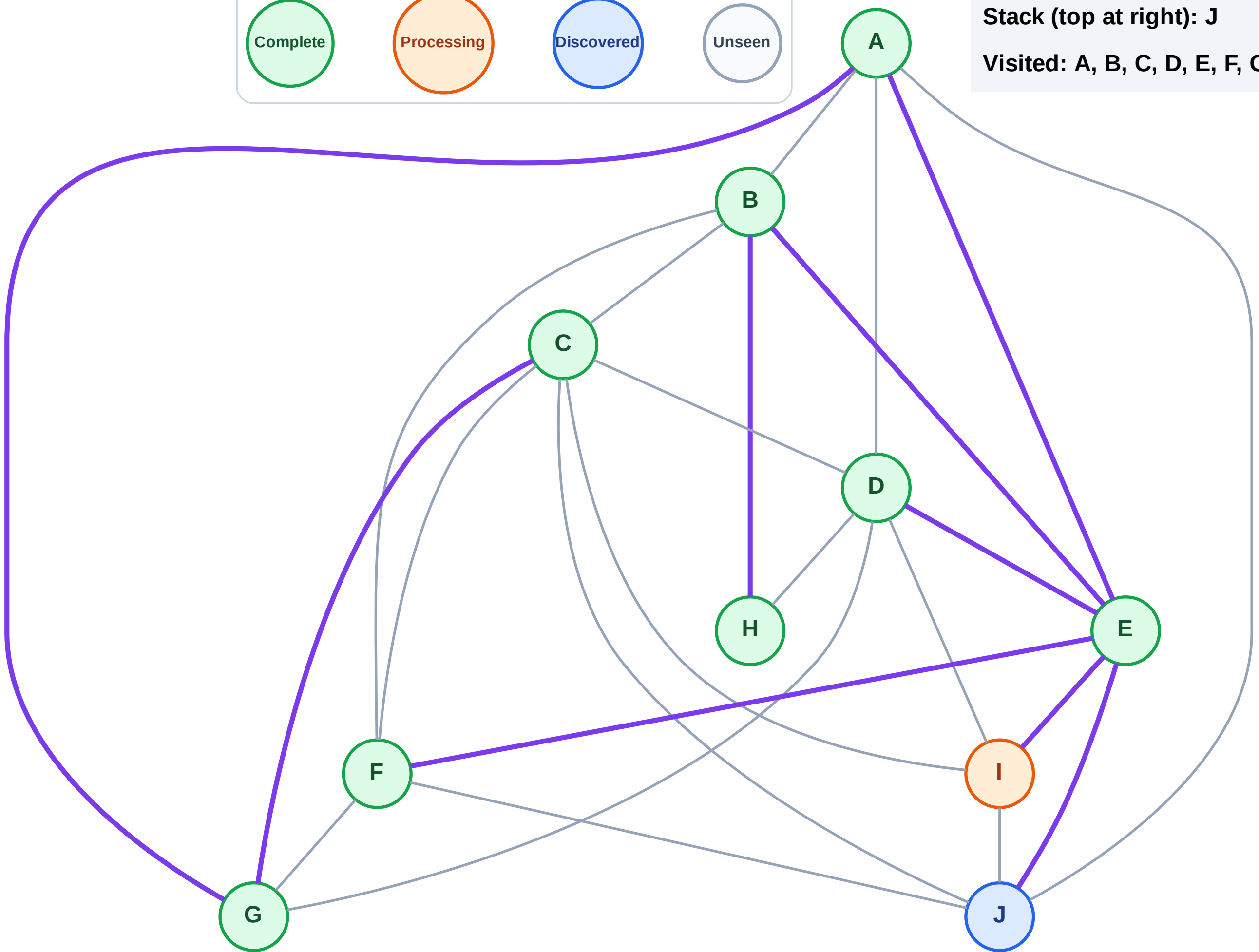
Step 18: Process node I

Legend



Stack (top at right): J

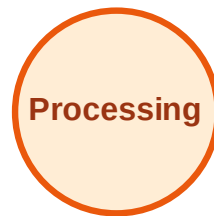
Visited: A, B, C, D, E, F, G, H



DFS Traversal

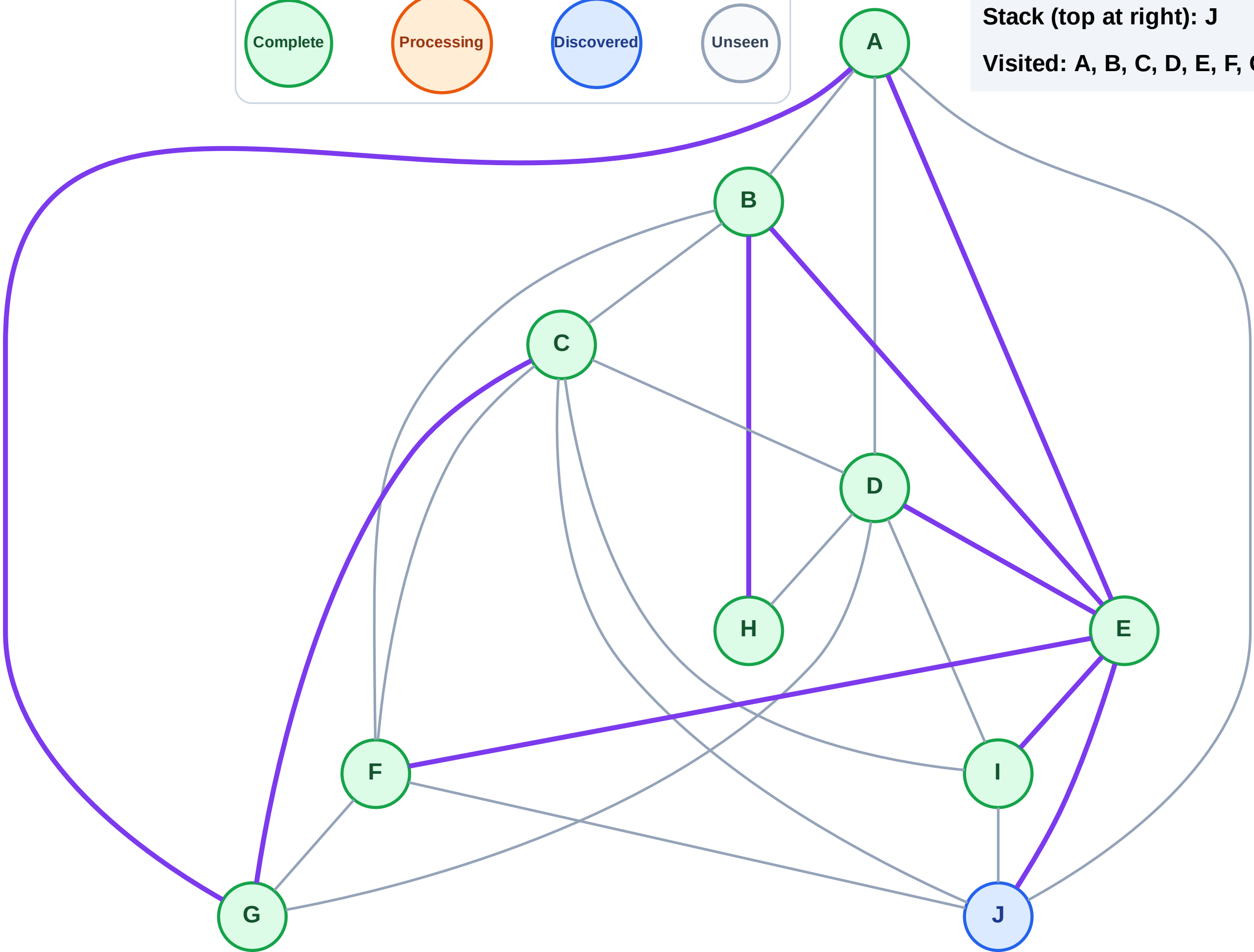
Step 19: No new neighbors from I

Legend



Stack (top at right): J

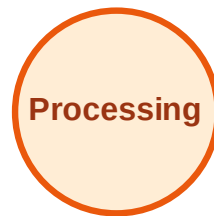
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

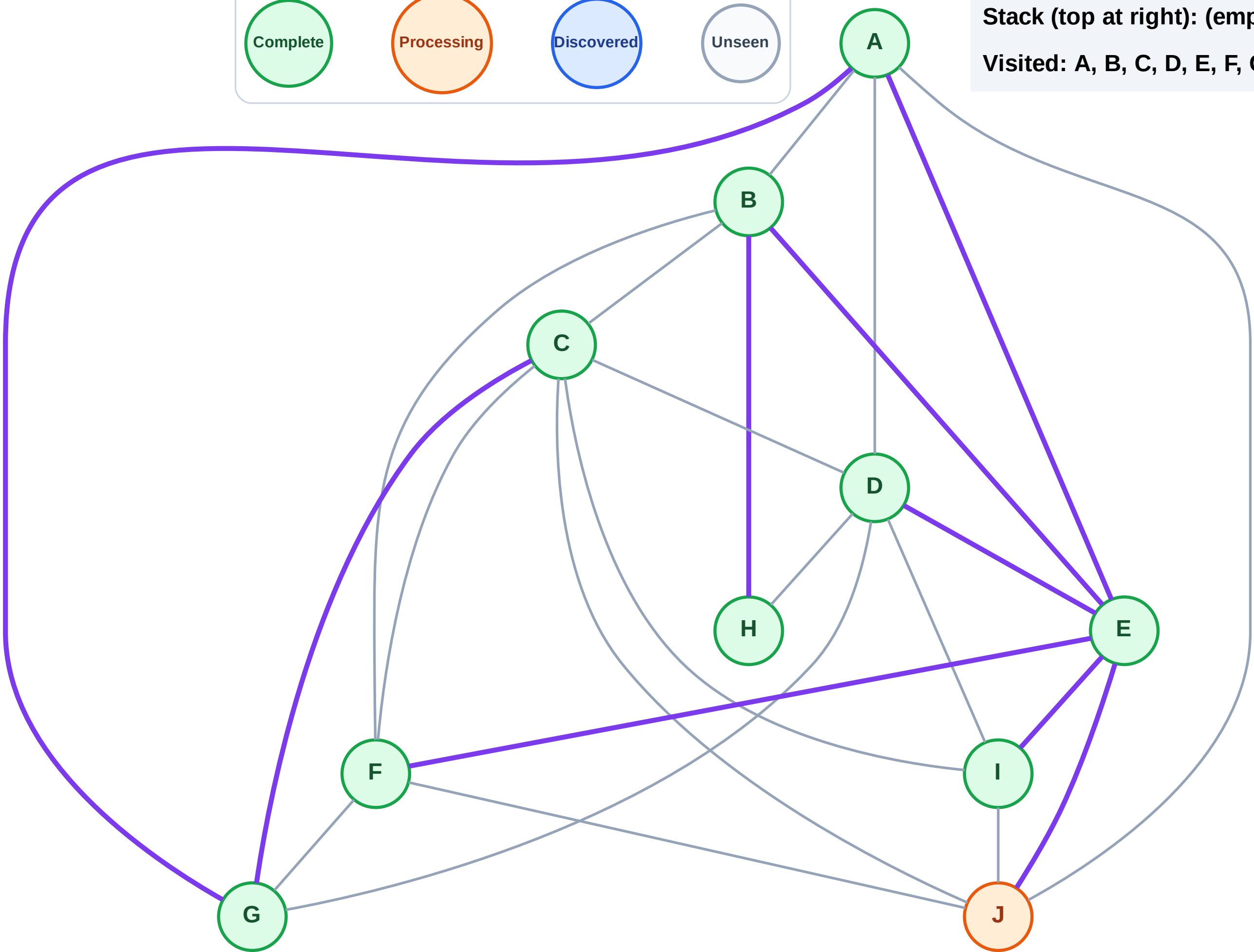
Step 20: Process node J

Legend



Stack (top at right): (empty)

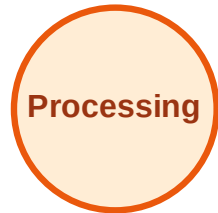
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

